Arithmetic Expression Geometry V: Triadic Characteristics and Threaded Apertures

Three Readings, Observation Fibres, and the Syntax of Apertures

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Abstract

This paper develops a theory of three coordinated typed readings of one carrier, temporal, spatial, and constructive, and of the syntax of the apertures through which they are filled and checked. A finite observer contract fixes an observation map, an equivalence, projectors, and a budget; a characteristic is then an operational quotient together with a residual, and the recurring theorem shape is negative: values, gradients, spectra, and energies do not recover history.

In the quadratic family $A_\kappa = \mathbb{R}[j]/(j^2 - \kappa)$ every member satisfies rank-one recovery, conjugation, and a multiplicative norm, and a globally defined negative-inverse involution, equivalently an oriented order-four lift with $J^2 = -I$ over $[J]^2 = [I]$, selects the complex member; chirality is the central sign that projectivization forgets. On a finite cut interval the six-state carrier $V \oplus \Lambda^2 V$ is not postulated: cut-membership cochains with oriented interchange generate the Clifford anticommutator, whereas interchange alone provably cannot. In the Legendre family one cusp is at once a branch-factor collision, a vanishing class carrying a Dehn transvection, and a period monodromy; a through-crossing written as a cospan with a resolution torsor derives $M_{\mathrm{Legendre}} = T_\delta^2$ and realizes the Picard–Lefschetz transvection on a finite cellular annulus. On the finite Legendre prism the total defect $(-I, [\infty, 1, 0])$ decomposes into two channels, so no irreducible triadic 3-cocycle is exhibited.

Part II takes the typed hole as primitive and proves threaded logic adequate on a finite terminal-world carrier under four explicit hypotheses. For the ordered linear fragment TND_0 it proves soundness and, through a grade invariant, incompleteness for Boolean semantics; an ordered one-hole substitution theorem; strong normalization, via ledger transport lifting through every context with two decreasing measures; a one-step strong diamond, hence local and global confluence and a unique normal form per fixed proof; the subformula property; and a terminating sound and complete focused decision procedure. A six-port whole-cut kernel answers "why six" by the counting bound $n \geq 2m$, hence $n_{\min} = 6$, under a stated contract, universality being weakened to relative initiality. Throughout, a quotient is admissible only with a declared residual and a calibration is never promoted to a theorem; triadicity is certified, not assumed.

Keywords: arithmetic expression geometry; triadic characteristic; observation fibre; operational quotient; residual; chirality; oriented exchange lift; Clifford relation; through-crossing; resolution torsor; monodromy; satisfaction form; aperture; typed hole; linear logic; strong normalization; confluence; focused derivability; six-port kernel

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Part I
Triadic characteristic geometry

1 Introduction and programme

1.1 The question this paper asks

Papers 0–IV of this series follow one expression through four successively coarser stages: an expression tree, a marked history, the operator that the history induces, and the endpoint value where it lands [3, 4]. Paper IV then asks when a quotient of that data is still sufficient for future continuation, and separates a semantic residual from an operational trace [5].

This paper asks a different question, which the earlier papers deliberately do not answer. Suppose one and the same carrier is to be read in three coordinated ways — say as a process, as a space, and as a construction — and suppose each reading is a typed interpretation with its own direction of information flow. What is the minimal structure that lets such readings be compared at all? What is necessarily lost in the comparison, and how is that loss to be accounted for? And what is the syntax in which the places of loss — the places where a reading is *not yet* determined — can be written down, filled, and checked?

The answer developed here has two halves, and they are the two parts of the paper.

1.2 Part I in one paragraph

Part I develops a theory of *triadic characteristics*. Its primitive is not a value but an *observer contract*: a finite datum fixing what is seen, when two things are declared indistinguishable, which projections are available, and what budget is granted. A characteristic of a process is then an operational quotient of that process together with a residual, and the residual is not a bookkeeping afterthought: it is the part of the construction that the quotient cannot see.

Given that framing, several classical-looking questions become sharp.

- Among the one-parameter quadratic scalar families $A_\kappa = \mathbb{R}[j]/(j^2 - \kappa)$, which one does the geometry force? The answer is that nothing in rank-one characteristic recovery, conjugation, or a multiplicative quadratic norm forces any of them; what selects $\mathbb{C}$ is a global negative-inverse involution on the punctured carrier, and this is equivalent, under one explicit bridge hypothesis, to the existence of an oriented order-four exchange lift J satisfying $J^2 = -I$ whose projective class satisfies $[J]^2 = [I]$. *Chirality* is precisely the central sign that projectivization forgets.
- Where does the six-state carrier $V \oplus \Lambda^2 V$ come from? Not from a postulate: on a declared finite cut interval, cut-membership cochains together with oriented interchange generate the Clifford anticommutator, whereas interchange alone provably cannot.
- What survives when one finite characteristic algebra is read temporally, spatially, and constructively? A stable variance signature, inference as constrained intersection of observation fibres, and a hard separation between four numeric invariants that a naive theory conflates.
- What is the right notion of *crossing*? Not a transport arrow, but a cospan with a resolution torsor; monodromy is then the difference of two resolutions, and the Legendre monodromy is *derived* from the quadratic base change rather than inserted by hand.

Part I also contains the paper's principal *negative* calibration. On the finite Legendre prism the total defect decomposes into a central sign channel and a history-suffix channel; therefore no

irreducible triadic 3-cocycle is exhibited there. We report this as a failure of the finite model, not as a success in disguise.

1.3 Part II in one paragraph

Part II takes seriously the place where a reading is undetermined. It makes the typed hole — the *aperture* — the primitive: syntactically empty, semantically a family of fillings, and with open and close operations that are deliberately not inverse.

On that primitive it builds a finite semantics (a terminal-world carrier, filling fibres, and an adequacy theorem for threaded logic under explicit hypotheses), an ordered linear natural-deduction fragment that is sound but provably incomplete for Boolean semantics, and a proof theory strong enough to be interesting: an ordered one-hole substitution law with a ledger splice, conditional fresh-cut admissibility, strong normalization via two decreasing measures, local and global confluence with a unique normal form per fixed proof, and finally a focused normal-form characterization with the subformula property that yields a terminating, sound, and complete decision procedure for derivability.

Part II closes with the six-port whole-cut kernel, which answers the question "why six ports?" by a counting theorem $n \geq 2m$ under a stated contract, and which weakens universality to relative initiality rather than claiming it.

1.4 The discipline the two parts share

Three rules govern every claim in this paper, and the reader is invited to hold the paper to them.

1. **A quotient is admissible only with a declared residual.** Every quotient result must either be accompanied by an exact reconstruction statement or name a residual whose contents are characterized. The phrase "the quotient loses nothing important" is not an available move in this paper.

2. **A calibration is never promoted to a theorem.** Where a statement is established only by an exhaustive check of a declared finite fixture, it is labelled a calibration, and the fixture is named. Where it is established by an argument, the argument appears in the paper or in an appendix.

3. **A property is certified, not named.** Three readings are triadic only if the pairwise crossings are given and the cyclic residual is exhibited. Universality is only relative initiality unless a stronger theorem is supplied. A boundary-return word is not a periodic orbit.

These rules are the reason Part I contains as many no-go results as positive ones. They are also the reason Part II spends effort on metatheorems: soundness alone is cheap, and the interesting question is always what fails.

1.5 Reading guide and dependencies

A reader interested only in the geometry of readings may read Part I, taking §2–§5 as the core and §7–§8 as the calibrations that constrain it. A reader interested in the syntax of apertures may read Part II from §11, taking §13–§15 as its proof-theoretic content. The two parts meet at the notion of a filling fibre: in Part I a residual is what a quotient cannot see, and in Part II an aperture is where a partial reading records that it cannot yet see.

The paper imports, and does not redefine:

- from Paper 0, the hyperbolic model, the affine cocycle and its relative defect, the Möbius involution $z \mapsto -1/z$, and the cut-and-glue language [3];
- from Paper I, the separation of literal syntax, marked history, induced operator, and endpoint value, and the projective-versus-affine distinction [4];
- from Paper IV, the requirement that a quotient be judged against a declared continuation class and observation, and the separation of a semantic bound from an operational trace [5].

1.6 Conventions and notation

Throughout, $\mathbb{R}$ and $\mathbb{C}$ are the real and complex fields, $\mathbb{F}_2$ the field with two elements, and $\mathbb{P}^1$ the projective line. For a finite set X we write $\mathcal{P}(X)$ for its powerset and $\mathbf{2} = \{\bot, \top\}$ for the Sierpinski lattice with $\bot < \top$.

An *observer* is written Q; its components are introduced in §2 and are never abbreviated away. A *characteristic* is written χ, and $\mathrm{Fib}_{D,Q}(x)$ denotes an observation fibre; the subscript always names both the domain and the observer, because omitting either is the most common source of error in this material.

Crossings are written with subscripts naming the two readings being compared, so that $J_{\mathcal{K}\mathcal{X}}$ and $J_{\mathcal{X}t}$ are different maps and composition order matters. The orientation expression of the six-state carrier is Ω, and it is a *derived* object (§4) whenever the derivation is available.

In Part II, a hole is written h_d with d its domain slot; $\mathrm{Fill}_{\mathcal{M},\rho}$ is the filling family in model $\mathcal{M}$ under assignment ρ; TND_0 names the ordered linear natural-deduction fragment of §13. The dedicated environments Calibration, Proposal, and Conjecture carry the epistemic weight that the words suggest, and a Theorem or Proposition is never used for a finite fixture check.

1.7 What this paper does not establish

For emphasis, and because the material is exploratory in places, the following are *not* claimed anywhere in this paper.

- That the three named readings are canonically identified. They are compared through declared crossings with declared residuals; a canonical identification is an open problem (§17).
- That arithmetic programs must use complex foundational scalars. Only that, within the tested quadratic family and under a stated bridge hypothesis, one property selects $\mathbb{C}$ (§3).
- That the six-state carrier is a spacetime decomposition or a physical claim of any kind. The derivation is finite and combinatorial (§4).
- That any finite calibration demonstrates universality, non-computability, or a physical realization.
- That order can be inferred from an energy, a grade, or a scalar. The opposite is proved in several places (§9, §10).
- That the triadic programme has been completed. Its decisive missing theorem is named in §17.

2 Observers, fibres, and the residual principle

Everything in this paper is relative to what some observer is allowed to see. This section fixes that relativity once, so that later statements about "what cannot be recovered" have a definite scope. The material is developed on one deliberately small worked model, the binary affine monoid; nothing in §5 or later depends on that model being the only one available.

2.1 The observer contract

Definition 2.1 (Observer contract)**.** Let $\mathcal{D}$ be a class of processes. A *finite observer contract* on $\mathcal{D}$ is a tuple

$$Q = (F_Q, \equiv_Q, \Pi_Q, B_Q)$$

consisting of

1. an *observation map* $F_Q \colon \mathcal{D} \to \mathcal{O}_Q$ into a codomain of readable data, possibly many-sorted;
2. an *equivalence* $\equiv_Q$ on $\mathcal{O}_Q$, declaring when two readable data are indistinguishable to Q;
3. a family of *projectors* Π_Q, each specifying which part of an observation is retained by a given reading;
4. a *budget* B_Q, bounding the finite computations Q may perform.

The budget is part of the contract, not an implementation detail. A claim of the form "no observer of class $\mathcal{O}$ can determine X" has no content unless $\mathcal{O}$ is fixed, and B_Q is where the finiteness of the observation enters.

Remark 2.2. The four components are not independent. Enlarging a budget can refine $\equiv_Q$, and adding a projector can shrink F_Q's image. Every statement below that quantifies over observers therefore names the contract it uses.

2.2 Fibres and characteristics

Definition 2.3 (Observation fibre)**.** For $o \in \mathcal{O}_Q$ and a reading $D \in \{t, X, \mathcal{K}\}$ (temporal, spatial, constructive), the *fibre* of o under D is

$$\mathrm{Fib}_{D,Q}(o) = \{\, p \in \mathcal{D} : F_{D,Q}(p) \equiv_Q o \,\}.$$

Definition 2.4 (Characteristic)**.** A *characteristic* of p for the contract Q is a triple

$$\chi_Q(p) = (\bar{\chi}_Q(p),\ \pi_p,\ R_Q(p)),$$

where $\bar{\chi}_Q(p)$ is the operational quotient (the readable value), π_p is a certificate that p maps to it, and $R_Q(p)$ is the *residual*: the data retained about p that the quotient does not determine.

Remark 2.5 (A characteristic is not a value). Definition 2.4 is where this paper departs from the usual usage. A spectrum, a gradient, an eigenvalue, or a scalar is at best $\bar{\chi}_Q$; the residual is part of the object, and propositions below exhibit cases where discarding it destroys exactly the information a later reading needs.

2.3 The residual principle

The following principle is used as an admissibility test throughout; it is the Part I counterpart of the rule that no quotient may be used without a declared residual.

Proposal 2.6 (Residual principle)**.** A quotient $\bar{\chi}_Q$ of a class $\mathcal{D}$ may be used as a state for a declared class of continuations and observations only when accompanied by

1. an exact reconstruction statement for the continuations in question, or
2. a residual R_Q whose contents are characterized, together with a stated sense in which the continuation cannot depend on what R_Q omits.

This is a Proposal rather than a theorem because options (i) and (ii) are requirements on presentation, not mathematical assertions. Its bite is in what it forbids: a quotient presented without either is *not* eligible as a state.

2.4 Identifiability is relative to a hypothesis class

The worked model is the binary affine monoid. Let

$$f_{k,b}(x) = 2^k x + b, \qquad k, b \in \mathbb{Z},$$

with composition in execution order. Writing S for $f_{0,1}$ (successor) and D for $f_{1,0}$ (doubling), every element has a normal form $D^k; S^b$.

Lemma 2.7 (Affine product law)**.** *For $f_i = f_{k_i,b_i}$ executed in the order f_1 then f_2,*

$$f_2 \odot f_1 = \big(k_1 + k_2,\ 2^{k_2} b_1 + b_2\big),$$

with identity $f_{0,0}$. In particular

$$\chi(S; D) = (1, 2) \neq (1, 1) = \chi(D; S).$$

Proof. $f_2(f_1(x)) = 2^{k_2}(2^{k_1}x + b_1) + b_2 = 2^{k_1+k_2}x + 2^{k_2}b_1 + b_2$. The two composites differ in their second coordinate because $2 \cdot 1 + 0 = 2$ while $2^0 \cdot 0 + 1 = 1$. □

Identifiability of (k, b) is not an absolute property; it is relative to a declared class of hypotheses and to the probes the contract allows.

Proposition 2.8 (Two-probe temporal recovery)**.** *Probing the temporal reading at $x = 0$ and $x = 1$ recovers (k, b) uniquely within the affine class: with $y_i = f_{k,b}(i)$ one has $b = y_0$ and $2^k = y_1 - y_0$.*

Proof. Immediate from $f_{k,b}(0) = b$ and $f_{k,b}(1) = 2^k + b$. □

Remark 2.9 (One probe is not enough)*.* A single probe at $x = 1$ cannot separate S from D, since $f_{0,1}(1) = 2 = f_{1,0}(1)$. The two-probe result is therefore a statement about the contract, not about the process.

The same phenomenon appears on the spatial side, and the contrast is instructive: there a single *exact interval* suffices, whereas a radius alone does not.

Proposition 2.10 (Spatial recovery from one exact interval)**.** *Let the spatial reading act on closed intervals by inverse image, $\rho_X(k, b)(I(m, r)) = I\big((m - b)/2^k,\ r/2^k\big)$. From the single observation $I(m, r) \mapsto I(m', r')$ one recovers $2^k = r/r'$ and hence $b = m - 2^k m'$.*

Proof. The observed image has midpoint $(m - b)/2^k$ and radius $r/2^k$. Dividing gives 2^k, and substituting gives b. □

Warning 2.11*.* Proposition 2.10 does not say that radii determine translations. With the radius alone the observation is $r \mapsto r/2^k$, which is blind to b entirely. The informative observer is doing the work, not the mathematics of affine maps.

2.5 The four-valued outcome, and what "unknown" means

Inference in this setting is not a function into answers; it is a computation that may fail to conclude. We therefore insist on four outcomes.

Definition 2.12 (Outcome discipline)**.** An inference query returns exactly one of *unique*, *ambiguous*, *inconsistent*, or *unknown*. An empty candidate set is reported as *inconsistent* only when exhaustion of the declared search space is certified; a budget expiry or an incomplete enumeration is reported as *unknown*.

Warning 2.13 (Failure to find is not a proof of absence). Definition 2.12 forbids the inference "search found nothing, therefore nothing exists". This prohibition recurs in Part II, where an exhausted search and a certified empty fibre are deliberately distinguished (§12).

2.6 The quotient is lossy, and the loss is the point

It is tempting to treat a characteristic as a compressed description of a process. The affine model shows immediately that compression and loss are the same phenomenon.

Proposition 2.14 (Quotient without section)**.** *Let $\chi_{\mathcal{K}}$ be the constructive characteristic that records the parameter pair of a normal form. Then*

$$\chi_{\mathcal{K}}(S; D) = \chi_{\mathcal{K}}(D; S; S) = (1, 2),$$

so $\chi_{\mathcal{K}}$ does not distinguish the raw histories, although $\chi_{\mathcal{K}} \circ \mathrm{NF} = \mathrm{id}$ holds on parameter pairs. Consequently $\mathrm{NF} \circ \chi_{\mathcal{K}} \neq \mathrm{id}$ on raw histories.

Proof. The composites are $f_{1,0} \circ f_{0,1} = f_{1,2}$ and $f_{0,1} \circ f_{0,1} \circ f_{1,0} = f_{1,2}$ by Lemma 2.7, while the words $S; D$ and $D; S; S$ are distinct. □

Proposition 2.15 (Compression is inseparable from loss)**.** *The characteristic of $S; D^n$ is the pair $(n, 2^n)$, whose canonical word $D^n; S^{2^n}$ has 2^n successor symbols, while the parameter pair requires $O(n)$ bits. The compression ratio is therefore exponential, and it is purchased exactly by the identification in Proposition 2.14.*

Proof. By induction on n using Lemma 2.7: $f_{n,2^n} \odot S = f_{n,2^n+1}$ and doubling increments the first coordinate. The normal form of $(n, 2^n)$ is $D^n; S^{2^n}$ by construction. □

Remark 2.16. Propositions 2.14 and 2.15 are the smallest instance of the pattern that organises the whole paper: a quotient is useful in proportion to what it forgets, and admissible in proportion to whether the forgotten part is accounted for.

2.7 Summary of what this section fixes

The section has introduced the observer contract, the fibre, the characteristic with its residual, and the outcome discipline. Two things should be carried forward.

First, *identifiability is a property of a contract*, not of a process (Proposition 2.8 and Remark 2.9).

Second, *a quotient must be presented with a residual* (Proposition 2.6 and Proposition 2.14). Every later section that computes a characteristic states which of the two options in Proposition 2.6 it uses.

3 The quadratic trichotomy and the central sign

A recurring temptation in this material is to declare the complex field as the foundational scalar and then to derive the geometry. This section shows why that order is unavailable. Within the natural one-parameter family of two-dimensional real algebras, the coarse requirements that a "characteristic" theory might impose are satisfied by all three members; only a genuinely geometric condition separates them, and that condition has a precise meaning: it is the retention of a central sign.

3.1 The quadratic family

Definition 3.1 (Quadratic scalar family)**.** For $\kappa \in \{-1, 0, 1\}$ let

$$A_\kappa = \mathbb{R}[j]/(j^2 - \kappa),$$

so that $a + bj$ with $a, b \in \mathbb{R}$ multiplies by

$$(a + bj)(c + dj) = (ac + \kappa bd) + (ad + bc)j.$$

Proposition 3.2 (Elementary structure)**.** *In each A_κ: the quadratic norm $N_\kappa(x) = x\bar{x}$ with $\overline{a + bj} = a - bj$ is multiplicative, and conjugation is a multiplicative involution. Moreover for every $u \in A_\kappa$ the rank-one map $L_u(x) = ux$ satisfies*

$$L_u(1) = u, \qquad L_u \circ L_v = L_{uv},$$

so a scalar is recovered from its own characteristic action without coordinates, and the three aspect units $\tau, \sigma, \rho \in A_\kappa$ compose cyclically to a single characteristic $L_{\rho\sigma\tau}$.

Proof. Multiplicativity of N_κ is the identity $(ac + \kappa bd)^2 - \kappa(ad + bc)^2 = (a^2 - \kappa b^2)(c^2 - \kappa d^2)$, which is a direct expansion. The rest is immediate from associativity of A_κ and $L_u(1) = u$. □

The three cases are conventionally named by the sign of the norm:

κ	algebra	$N_\kappa(a, b)$	type
-1	complex numbers	$a^2 + b^2$	elliptic
0	dual numbers	a^2	parabolic
1	split-complex numbers	$a^2 - b^2$	hyperbolic

Remark 3.3 (What Proposition 3.2 does not do). It does not select $\kappa = -1$. Rank-one recovery of a characteristic from its own action, multiplicative conjugation, and a multiplicative quadratic norm are all satisfied by the three members. Any argument that begins "the scalar must be complex, since only then can one conjugate and take norms" is therefore invalid.

3.2 The three nondegeneracy conditions, and why they are not derivations

Three further requirements do separate the cases.

Proposition 3.4. *Within $\{A_\kappa\}$ each of the following conditions holds for $\kappa = -1$ alone:*

1. *every nonzero element is invertible;*
2. *N_κ is positive definite;*

3. the unit-norm group $\{x : N_\kappa(x) = 1\}$ *is compact.*

Proof. For $\kappa = -1$ the inverse is $(a + bj)^{-1} = (a - bj)/(a^2 + b^2)$ and the unit group is S^1. For $\kappa = 0$, every bj with $b \neq 0$ is a noninvertible nonzero element. For $\kappa = 1$, $(1 + j)(1 - j) = 0$ exhibits nonzero zero divisors and the norm is indefinite with noncompact unit locus. □

Warning 3.5. Proposition 3.4 is a separation result, not a derivation. None of the three conditions is forced by the geometry of readings. They are *available* selection principles, and choosing one is a mathematical decision that must be declared.

3.3 The negative-inverse criterion

A better candidate comes from the geometry itself. Paper 0 already contains the Möbius involution $S(z) = -1/z$ as the reciprocal reading [3]; the question is which quadratic scalars admit it globally.

Proposition 3.6 (Negative inverse in the quadratic family)**.** *For* $z = a + bj \in A_\kappa$ *with* $a^2 - \kappa b^2 \neq 0$*,*

$$S_\kappa(a, b) = \Big(\frac{-a}{a^2 - \kappa b^2},\ \frac{b}{a^2 - \kappa b^2}\Big)$$

satisfies $z\, S_\kappa(z) = -1$*. For* $\kappa = -1$ *the formula*

1. *is defined on the whole punctured plane* $\mathbb{R}^2 \setminus \{0\}$*;*
2. *is an involution;*
3. *satisfies* $zS_{-1}(z) = -1$ *identically;*
4. *preserves the upper half-plane* $\{b > 0\}$*.*

For $\kappa = 0$ *the entire nilpotent axis* $a = 0$ *is excluded. For* $\kappa = 1$ *the two light rays* $a = \pm b$ *are excluded, and the sign of the second coordinate need not be preserved across the components they separate.*

Proof. $1/z = (a - bj)/(a^2 - \kappa b^2)$, so $-1/z = S_\kappa(a, b)$ as displayed, and $zS_\kappa(z) = -1$ follows. For $\kappa = -1$ the denominator $a^2 + b^2$ never vanishes off the origin, and $b/(a^2 + b^2) > 0$ whenever $b > 0$; involutivity is the usual $(S \circ S)(z) = z$ for a Möbius involution on a chart where it is defined. The excluded loci for $\kappa = 0, 1$ are exactly the zero divisors. □

Proposal 3.7 (The AEG inversion criterion)**.** Among the tested family, $\mathbb{C}$ is selected by the requirement that a globally defined negative-inverse involution exist on every nonzero scalar and preserve one connected real half-plane.

This is labelled Proposal because the requirement is imported from the observed transition between two real grids, not derived from program geometry; what is proved is only that it selects.

3.4 The oriented lift, and chirality as a central sign

The inversion criterion has a sharper form which explains *why* it selects, and which supplies the paper's working definition of chirality.

Definition 3.8 (Time–space exchange and its lift)**.** On a homogeneous pair (t, s) put

$$J(t, s) = (-s, t), \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

On the projective chart $z = t/s$ this induces $[t : s] \mapsto [-s : t]$, i.e. $z \mapsto -1/z$. We call J the *oriented lift* of the projective exchange.

Proposition 3.9 (Order two versus order four)**.** $[J]^2 = [I]$ *on the projective line, while* $J^2 = -I$ *on the oriented lift. Hence* $J^{-1} = -J$*, and the forward and reverse exchanges induce the same projective involution although they are distinct on the lift.*

Proof. $J^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Projectively $[-\mathrm{id}] = [\mathrm{id}]$, so $[J]^2 = [\mathrm{id}]$, while on the lift the scalar -1 is retained. □

Definition 3.10 (Chirality)**.** *Chirality* is the central sign distinguishing the forward and reverse oriented lifts of one projective exchange: the datum that J and $J^{-1} = -J$ are different lifts of the same projective involution.

This definition is deliberately austere. It says nothing about the orientation of the ambient plane, and the following computation shows that it must not.

Proposition 3.11 (Chirality is not ambient orientation reversal)**.** *In real coordinates* $S(x, y) = (-x/(x^2 + y^2),\, y/(x^2 + y^2))$ *has Jacobian determinant*

$$\det DS = \frac{1}{(x^2 + y^2)^2} > 0.$$

Thus the involution is holomorphic and orientation-preserving on the plane; the chirality change is not a reversal of that plane but the difference between the two lifts of one projective map.

Proof. S is the Möbius map $-1/z$ with derivative $1/z^2$, so $\det DS = |z|^{-4} = (x^2 + y^2)^{-2} > 0$. □

3.5 The generator action under one bridge hypothesis

The central sign becomes a selection mechanism for the algebra when the same oriented two-dimensional carrier is asked to represent both the exchange lift and the action of the quadratic generator.

Proposition 3.12 (Generator action)**.** *Let* L_j *be multiplication by* j *on* (a, b)*, so that*

$$L_j = \begin{pmatrix} 0 & \kappa \\ 1 & 0 \end{pmatrix}, \qquad L_j^2 = \kappa I.$$

Then:

1. *for* $\kappa = -1$*,* $L_j = J$ *is orientation-preserving, invertible, of order four before projectivization;*
2. *for* $\kappa = 1$*,* $L_j^2 = +I$*,* $\det L_j = -1$*, and the generator is already its own inverse on the lift;*
3. *for* $\kappa = 0$*,* L_j *is singular and nilpotent.*

Proof. Direct computation for each κ; $\det L_j = -\kappa$ and $L_j^2 = \kappa I$. □

Calibration 3.13 (The bridge is a hypothesis)**.** Requiring an oriented time–space exchange whose projective class has order two while its lift distinguishes forward from reverse selects $j^2 = -1$, *provided* the same oriented two-dimensional carrier is declared to represent both the exchange lift and the generator action. Both halves are calibrated, not proved: the selection is a finite algebraic check, and the bridge is a declared identification.

Warning 3.14. The matrix $J \in SL(2, \mathbb{R})$ already defines the real-coefficient Möbius map $-1/z$; complex notation is therefore not by itself evidence that the underlying scalars are complex. What is new is that retaining the lift rather than quotienting to $\mathrm{PSL}(2, \mathbb{R})$ exposes the central sign. Calling that sign chirality requires the oriented carrier to remember it.

3.6 The conjugated presentation

The last piece of the trichotomy is what the involution does to the two AEG generators, because it shows the central sign in operational form.

Proposition 3.15 (Conjugated generators)**.** *Write $A(z) = z + 1$ and $M_\lambda(z) = \lambda z$ for $\lambda > 0$ in the upper half-plane chart. Then*

$$SM_\lambda S = M_{\lambda^{-1}}, \qquad SAS(z) = \frac{z}{1 - z}.$$

Thus conjugation inverts the scale direction while turning the translation grid into a parabolic grid organised around the other fixed point. For a Baumslag–Solitar-type relation $MAM^{-1} = A^n$ the conjugated presentation has scale generator M^{-1} and ripple generator SAS.

Proof. For the first identity, $SM_\lambda S(z) = -1/(\lambda \cdot (-1/z)) = z/\lambda = M_{\lambda^{-1}}(z)$. For the second, $SAS(z) = -1/((-1/z) + 1) = z/(1 - z)$, which is parabolic with the single fixed point $1/2$. □

Remark 3.16 (Baumslag–Solitar background). The relation $MAM^{-1} = A^n$ is the Baumslag–Solitar presentation [1]; Paper 0 records the exact routing of copy/branch/recombine programs through these relations. The content here is only that conjugation exchanges the forward and reverse presentations of the selected scale direction, which is the visible consequence of lift-level chirality.

3.7 What the trichotomy settles, and what it leaves open

Calibration 3.17 (Bounded red-team calibration)**.** Within the tested quadratic family: two real dimensions per aspect, cyclic rank-one action, recovery of a characteristic from its unit action, conjugation, and a multiplicative norm do *not* select $\mathbb{C}$; a global negative-inverse involution preserving a connected real AEG half-plane *does* select $\mathbb{C}$; and the mechanism is that the projective exchange has order two while its orientation-preserving lift has order four.

Three interpretations of the six real degrees of freedom remain live and are not separated by anything proved here.

Proposal 3.18 (Three readings of $6 = 2 + 2 + 2$)**.** The cases $\kappa = -1, 0, 1$ may respectively encode phase/orientation, infinitesimal extension, and causal scale; alternatively the three pairs may be three complex lines; alternatively $\mathbb{C}$, dual, and split behaviour may be charts or strata of one expression-valued scalar geometry. No canonical assignment of the three cases to temporal, spatial, and relational roles is established.

Remark 3.19 (The Jordan shoulder). The dual case is not a pathology to be eliminated. A nilpotent extension naturally records a first-order shoulder that survives complexification, which suggests that a faithful spectrum may require both a semisimple character and a residual extension object. §10 returns to this.

4 The six-state carrier and the derived Clifford relation

Three readings, each two-dimensional over the reals, give six real dimensions. The temptation is to equip each pair with a complex structure and declare the result $\mathbb{C}^3$. This section shows that the correct carrier is not a sum of three copies of anything: it is the six-dimensional space

$$W = \Lambda^1 E \oplus \Lambda^2 E, \qquad \dim_{\mathbb{R}} E = 3,$$

in which each reading is paired with the *oriented relation of the other two*, and in which the complex unit appears exactly once. The section then shows that on a declared finite cut interval this structure is not postulated but generated.

4.1 The carrier and its paired planes

Definition 4.1 (Six-state carrier)**.** Let E be a three-dimensional real vector space with basis T, S, R representing the temporal, spatial, and relational readings. Put

$$W = \Lambda^1 E \oplus \Lambda^2 E = \langle T, S, R\rangle \oplus \langle SR, RT, TS\rangle,$$

so $\dim_{\mathbb{R}} W = 6$. The three *aspect–opposite-relation planes* are

$$W_T = \langle T, SR\rangle, \qquad W_S = \langle S, RT\rangle, \qquad W_R = \langle R, TS\rangle.$$

The pairing is the content of the definition: a reading is not paired with a second copy of itself but with the oriented plane spanned by the other two. This is forced by the dimension count once one asks that each reading be recoverable from its own opposite relation, and it has an immediate consequence for the complex structure.

Proposition 4.2 (One complex unit, not three)**.** *Suppose the generators satisfy*

$$T^2 = S^2 = R^2 = 1, \qquad \varepsilon_i\varepsilon_j + \varepsilon_j\varepsilon_i = 0 \quad (i \neq j),$$

and set $\Omega = TSR$. Then Ω is central and

$$\Omega^2 = -1.$$

Consequently $J = \Omega\cdot$ preserves W, swaps $\Lambda^1 E$ with $\Lambda^2 E$ by cyclic pairing

$$J(T) = SR, \quad J(SR) = -T; \qquad J(S) = RT, \quad J(RT) = -S; \qquad J(R) = TS, \quad J(TS) = -R,$$

and on each plane W_T, W_S, W_R restricts to $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Hence $\chi_J(\lambda) = (\lambda^2 + 1)^3$.

Proof. Centrality. A generator T commutes with SR up to $(-1)^2 = +1$, so Ω lies in the centre.

The square. Moving the fourth factor of $\Omega^2 = T\,S\,R\,T\,S\,R$ to the front costs the two transpositions past R and S, hence $(-1)^2 = +1$:

$$T\,S\,R\,T\,S\,R = T^2\,S\,R\,S\,R = S\,R\,S\,R,$$

using $T^2 = 1$. Moving the third factor past the fourth costs one transposition:

$$S\,R\,S\,R = -\,S^2R^2 = -1.$$

Hence $\Omega^2 = -1$.

Pairing. For the first plane, $RT = -TR$ gives

$$\Omega T = T\,S\,R\,T = T\,S\,(-T\,R) = -(T\,S\,T)R = (T\,T\,S)R = T^2SR = SR,$$

and then $\Omega(SR) = T\,S\,R\,S\,R = -T\,S^2R^2 = -T$. The other two planes follow by cyclic symmetry. On the plane spanned by T and SR the action is therefore $T \mapsto SR \mapsto -T$, which is the displayed matrix.

Characteristic polynomial. J acts as the same 2×2 rotation on each of the three planes, so $\chi_J(\lambda) = \prod_i \det(\lambda I - J|_{W_i}) = (\lambda^2 + 1)^3$. □

Calibration 4.3 (The algebraic form is an assumption)**.** Proposition 4.2 takes the anticommutation relations as *hypotheses*. It is a bounded algebraic calibration: it shows what the carrier does once such generators exist. §4.3 supplies the derivation.

Proposition 4.4 (Chirality sign law)**.** *For every permutation π of the three directions,*

$$P_\pi \, \Omega \, P_\pi^{-1} = \operatorname{sgn}(\pi) \, \Omega.$$

In particular exchanging temporal and spatial directions gives $\Omega \mapsto -\Omega = \Omega^{-1}$, so cyclic permutations preserve the orientation expression and transpositions reverse it.

Proof. P_π permutes the generators, hence multiplies the word TSR by the sign of the permutation. Invariantly: a transposition reverses the induced orientation of E, and Ω is the orientation class of E followed by the volume of $\Lambda^2 E$ under Ω's own duality. □

Remark 4.5. Proposition 4.4 is the carrier-level form of Definition 3.10: the central sign is exactly what a transposition produces, and it is invisible to any projective reading.

4.2 The signature trichotomy

If the three directions are granted arbitrary signs, the complex unit becomes conditional.

Proposition 4.6 (Signature trichotomy)**.** *With $T^2 = \varepsilon_T$, $S^2 = \varepsilon_S$, $R^2 = \varepsilon_R$ and $\varepsilon_i \in \{-1, 0, +1\}$,*

$$\Omega^2 = -\varepsilon_T \varepsilon_S \varepsilon_R,$$

so the carrier is elliptic when the product is positive, hyperbolic when negative, and parabolic when the product vanishes.

Proof. Each T^2 contributes a factor ε_T to the reduction performed in the proof of Proposition 4.2; the signs multiply. □

Warning 4.7 (Temporality is not a metric sign). A Lorentzian reading with exactly one negative square forces the split case. Hence semantic temporality must *not* be identified with a negative metric direction; doing so silently converts the trichotomy of §3 into an unannounced choice of $\kappa = 1$.

4.3 Deriving the generators from cuts

The material above is algebraic. The geometric content of this section is that on a declared finite cut interval the two operations needed are already present, so the Clifford relation is *derived*.

Construction 4.8 (Local cut interval)**.** Let U be a finite set of events carrying the completion order of a checked acyclic program, and let g_1, g_2, g_3 be three pairwise independent enabled events generating a local Boolean cut interval. Two operations are available on cut data:

1. *oriented interchange*, which exchanges two independent enabled events and records the sign of the exchange, giving endomorphisms ε_i on the cut cochains;
2. *cut contraction*, which reads whether an event belongs to a cut, giving ι_i determined by the membership pairing
$$dx_i(g_j) = \delta_{ij}.$$

Theorem 4.9 (Endogenous Clifford relation)**.** *On the local cut interval of Construction 4.8 the following hold.*

1. *Oriented interchange yields the exterior relations* $\varepsilon_i\varepsilon_j + \varepsilon_j\varepsilon_i = 0$ *for* $i \neq j$*.*
2. *With membership pairing* $dx_i(g_j) = \delta_{ij}$*, the operators* $c_i = \varepsilon_i + \iota_i$ *satisfy*
$$c_ic_j + c_jc_i = 2\delta_{ij}I.$$
3. *Hence* $\Omega = c_1c_2c_3$ *satisfies* $\Omega^2 = -I$*, and on* $W = \Lambda^1E \oplus \Lambda^2E$ *the orientation action has* $\chi_J(\lambda) = (\lambda^2 + 1)^3$ *with three aspect–opposite-face planes.*

Proof sketch and status. Items (i) and (ii) are the Chevalley construction of a Clifford algebra from an exterior algebra together with a contraction satisfying the duality $dx_i(g_j) = \delta_{ij}$; what the construction adds is that both operations are *supplied by the cut data* rather than assumed. Item (iii) is then Proposition 4.2 applied to the derived generators. The derivation is checked exhaustively on one local three-cube and is reported as a bounded endogenous derivation, not as a theorem about arbitrary cut intervals. □

Theorem 4.10 (Interchange alone is insufficient)**.** *Oriented interchange by itself yields only a nilpotent exterior volume: it produces* $\varepsilon_1\varepsilon_2\varepsilon_3$ *satisfying* $(\varepsilon_1\varepsilon_2\varepsilon_3)^2 = 0$*, and no complex structure. The contraction* ι_i *is indispensable.*

Proof. Wedge products of degree three in a three-dimensional exterior algebra have square zero because any repeated factor annihilates the product. Without a contraction there is no degree-lowering operation, so the Chevalley combination $c_i = \varepsilon_i + \iota_i$ cannot be formed. □

Remark 4.11 (Why this matters). Theorems 4.9 and 4.10 together answer the natural objection to §3: the complex structure is not smuggled in with the carrier. It is produced by two operations that the cut interpretation already needs, and the second of them is doing essential work.

4.4 Filtrations from a designated causal line

One further structure is available, and it is important not to over-read it.

Proposition 4.12 (Six-state filtration)**.** *Designating one causal line refines the six-state carrier as*
$$6 = 1 + 2 + 2 + 1.$$
The two readings $(1, 2, 3)$ *and* $(1, 3, 2)$ *obtained by regrouping the transverse pairs differ by an orientation-unit transfer and are* not *a chiral pair.*

Status. The count is elementary from Definition 4.1. The absence of chirality is the content: see Theorem 4.13. □

Theorem 4.13 (No-go: regrouping is not chirality reversal)**.** *Orientation-unit transfer changes block membership but not* Ω*, whereas an actual odd exchange of two transverse directions satisfies* $P\Omega P^{-1} = -\Omega$*. Hence the passage* $(1, 2, 3) \leftrightarrow (1, 3, 2)$ *is a change of filtered reading and not a chiral pair. Consequently an ordinary scalar spectrum cannot distinguish the two readings, while the filtration can.*

Proof. The two facts are verified separately: the transfer preserves the orientation word by Proposition 4.4 applied to an even permutation, while an odd exchange is a transposition and therefore reverses it. A scalar spectrum depends only on Ω^2 and hence cannot see the sign at all. □

Remark 4.14. Theorem 4.13 is a small result with a large moral, which recurs in §10: quantities built from squares, norms, or spectra are blind precisely to the central sign that Definition 3.10 isolates.

5 Triadic characteristic inference

§2 fixed what an observer is and what a characteristic costs. This section reads one finite characteristic algebra in three ways at once and asks what the three readings share. The answer has two parts: a variance signature that is stable under every calibration performed so far, and an inference procedure that is nothing more than the intersection of three fibres. Both are developed on the affine monoid of Lemma 2.7 and then calibrated on the square and the cube in §6.

5.1 The three readings

Definition 5.1 (Typed readings)**.** Let $\mathcal{A}$ be a class of processes closed under a composition $\odot$ read in execution order. A *typed reading family* on $\mathcal{A}$ consists of

1. a *temporal* reading ρ_t, covariant in execution, taking values in endomorphisms of the state space;
2. a *spatial* reading ρ_X, contravariant, taking values in inverse-image maps on a space of admissible regions;
3. a *constructive* reading $\rho_{\mathcal{K}}$, covariant modulo a declared canonicalisation $\equiv_{\mathrm{can}}$, taking values in construction data.

The three readings have different variance by design: a process acts forward on states, backward on regions, and forward again on constructions. This asymmetry is not a defect; it is what the phrase "three readings" is meant to encode.

Proposition 5.2 (Variance signature)**.** *Let f_1 be executed before f_2 and write $f_2 \odot f_1$ for the composite. Then*

$$\rho_t(f_2 \odot f_1) = \rho_t(f_2) \circ \rho_t(f_1), \qquad \rho_X(f_2 \odot f_1) = \rho_X(f_1) \circ \rho_X(f_2),$$

$$\rho_{\mathcal{K}}(f_2 \odot f_1) \equiv_{\mathrm{can}} \rho_{\mathcal{K}}(f_2) \circ \rho_{\mathcal{K}}(f_1).$$

Schematically, writing $\mathfrak{T}$ for the triple,

$$\mathfrak{T}(f_2 \odot f_1) = \big(T_2 \circ T_1,\ X_1 \circ X_2,\ K_2 \circ K_1 \bmod \mathrm{can}\big),$$

so the signature is $(+,-,+)$ in the order $(t, X, \mathcal{K})$.

Proof. The temporal and constructive statements are functoriality of ρ_t and of $\rho_{\mathcal{K}}$ modulo canonicalisation. The spatial statement is contravariance of the inverse-image construction: a region transformed by the composite f_1 then f_2 is $\rho_X(f_2)^{-1}\big(\rho_X(f_1)^{-1}(R)\big) = \big(\rho_X(f_1) \circ \rho_X(f_2)\big)^{-1}(R)$. □

Calibration 5.3 (Scope of the signature)**.** The signature of Proposition 5.2 is established by hand inside the finite affine model, and re-verified in the calibrations of §6 and §7. It is *not* claimed as a theorem about every conceivable three-domain model. What changes from model to model is the structure each reading must carry; the variance law has so far not changed.

5.2 Inference as intersection of fibres

Proposition 5.4 (Triadic inference)**.** *For observations $o_t, o_X, o_{\mathcal{K}}$ presented to the three readings, the inferred set of candidates is*

$$\mathcal{I}(o_t, o_X, o_{\mathcal{K}}) = \mathrm{Fib}_{t,Q}(o_t) \cap \mathrm{Fib}_{X,Q}(o_X) \cap \mathrm{Fib}_{\mathcal{K},Q}(o_{\mathcal{K}}).$$

Each reading contributes constraints and the constraints are intersected.

Proof. By Definition 2.3, a process lies in $\mathrm{Fib}_{D,Q}(o_D)$ exactly when its D-reading is Q-indistinguishable from o_D. A process satisfying all three observations therefore lies in the intersection, and conversely. □

Remark 5.5 (What "one algorithm" can and cannot mean). Proposition 5.4 says that the three readings share one *operation* in characteristic space. It does not say that they share an untyped decoder; the fibres live in different codomains and the intersection is taken after transport. Reading the proposition as a claim about a universal interpreter is a mistake, and it is the mistake that Part II is designed to prevent.

The outcome of the intersection is reported through Definition 2.12. In particular Warning 2.13 applies with force here: an empty intersection is *inconsistent* only if all three searches were exhausted within B_Q.

5.3 The deformed crossing

Inference is only half the story; the other half is how readings move past one another. The affine model gives an exact law, and it is not a swap.

Proposition 5.6 (Order reduction is not a swap)**.** *Write S for successor and D for doubling. Then*

$$S; D \equiv D; S; S, \qquad \textit{while} \qquad S; D \not\equiv D; S,$$

where $\equiv$ denotes extensional equality of the induced value. Moving D leftward across S duplicates *the translation generator, and every pair has a canonical representative $D^k; S^b$.*

Proof. By Lemma 2.7, $S; D$ has parameters $(1, 2)$ while $D; S$ has $(1, 1)$; and $D; S; S$ has $(1, 2)$ as well. Canonicality of $D^k; S^b$ is the normal form of the monoid. □

Warning 5.7 (Extensional, not confluent). Proposition 5.6 is one exact extensional crossing law. It is *not* a terminating and confluent rewriting theorem for the abstract presentation, and this paper does not claim one. The distinction matters because Part II proves genuine normalization theorems under linearity hypotheses, and Proposition 5.6 must not be mistaken for one of them.

5.4 The square map: loss, branching, and reverse transport

The clearest model of what a reading can and cannot retain is the real square.

Proposition 5.8 (Temporal loss is an orbit quotient)**.** *Let $q(x) = x^2$ and $\sigma(x) = -x$. Then $q \circ \sigma = q$, so q factors as*

$$\mathbb{R} \xrightarrow{\text{quot}} \mathbb{R}/C_2 \cong [0, \infty) \xrightarrow{\hookrightarrow} \mathbb{R}.$$

The generic fibre is a single two-element C_2-orbit; $\{0\}$ is a fixed point, so the quotient is non-free exactly at the origin.

Proposition 5.9 (Spatial branching is the same quotient, read backwards)**.** *On $\mathbb{R}^\times \to \mathbb{R}_{>0}$, q is a two-sheeted covering and σ exchanges the sheets. For $0 < a < b$,*

$$q^{-1}(a, b) = (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b}),$$

while $q^{-1}(-1, 1) = (-1, 1)$ exhibits the sheets merging through the critical source point.

Proposition 5.10 (Reverse transport is relation-valued)**.** *The backward reading $q^\dagger$ is $\varnothing$, $\{0\}$, or $\{-\sqrt{y}, \sqrt{y}\}$ according as $y < 0$, $y = 0$, or $y > 0$. Hence a plain global inverse function does not exist, although forward execution remains functional.*

Remark 5.11. Propositions 5.8–5.10 are the smallest complete instance of the warning that contravariance forces relation-valued transports. The spatial reading of a critical source is not a function, and any formalism that demands one will either lose the singularity or invent a branch convention that is not intrinsic to the data.

5.5 Scaling crosses; translation does not

Proposition 5.12 (Scaling crosses by deformation)**.** *For $D_a(x) = ax$ and $Q(x) = x^2$,*

$$D_a; Q \equiv Q; D_{a^2},$$

with equal extensional value and different raw histories.

Proof. $q(D_a x) = a^2 x^2$ and $D_{a^2}(q(x)) = a^2 x^2$. □

Theorem 5.13 (No-go: translation has no affine crossing)**.** *Let $S_b(x) = x + b$ with $b \neq 0$. There is no affine $A(y) = \alpha y + \beta$ with*

$$S_b; Q \equiv Q; A.$$

Proof. Every post-composite of Q with an affine map has the form $A(q(x)) = \alpha x^2 + \beta$, whose linear coefficient vanishes identically. But $q(S_b x) = x^2 + 2bx + b^2$ has linear coefficient $2b \neq 0$ since the field has characteristic 0 and $b \neq 0$. □

Remark 5.14 (Scope of Theorem 5.13). The theorem forbids an *affine* crossing. It does not forbid a non-affine expansion, and the next result exhibits exactly that. Reading the obstruction as a claim about all rewrites would overstate it; this is the sense in which the statement is a boundary statement about the affine post-normal-form language, not about reduction in general.

Proposition 5.15 (The word-to-graph boundary)**.** *Crossing a translation through a square forces a parallel decomposition with explicit copy:*

$$x \xrightarrow{\Delta_3} (x_1, x_2, x_3) \xrightarrow{Q \otimes D_{2b} \otimes C_{b^2}} (x_1^2,\ 2bx_2,\ b^2) \xrightarrow{\mathrm{add}_3} x^2 + 2bx + b^2.$$

Consequently general reduction in this setting is not a global ordering of two word alphabets, but typed graph rewriting with declared copy and merge.

Proof. The right-hand side of the obstruction is a sum of three terms whose coefficients come from three distinct uses of one source. A word over the one-input alphabets $\{S_b\}$ and $\{Q\}$ cannot express three simultaneous uses of its single input; a graph with an explicit fan-out can. □

Calibration 5.16 (Constructive fibre is not a singleton)**.** $\chi_{\mathcal{K}}(Q) = \chi_{\mathcal{K}}(N; Q) = q$ while the raw histories and the construction data of Q and $N; Q$ remain distinct. Hence the constructive fibre is not a singleton even when the temporal and spatial fibres are, and by Proposition 2.6 a usable characteristic must retain the residual.

5.6 Summary

Three results in this section are load-bearing for what follows.

1. The variance signature $(+, -, +)$ is stable across the models tested (Proposition 5.2, Calibration 5.3).
2. Crossings are deformations, and translation does not cross an affine normal form at all (Propositions 5.6, 5.12, Theorem 5.13).
3. The consequence is a change of carrier: from words to typed graphs with explicit copy (Proposition 5.15). Part II takes up precisely this carrier, where the copy discipline becomes a linearity discipline.

6 Typed multiplicity, equivariance, and the word-to-graph boundary

The temptation produced by §5 is to suppose that a reading has a single "multiplicity" — a degree, a number of preimages, a count of sheets — and that these agree. They do not. This section separates them, shows that the correct general form of the stability a characteristic enjoys is equivariance rather than invariance, and exhibits the coaction responsible for the failure of translation to cross a power inside a word normal form.

6.1 Four invariants, not one

Definition 6.1 (Observer-relative degree profile)**.** For a process P and a contract Q, the *degree profile* is the quadruple

$$\mathbf{d}_Q(P) = (d_{\text{alg}},\, d_{\text{fib}},\, d_{\text{sheet}},\, d_{\text{occ}}),$$

where d_{alg} is the algebraic degree of the induced characteristic, d_{fib} the cardinality of a generic fibre of the reverse reading over the declared scalar domain, d_{sheet} the covering degree of the associated spatial correspondence, and d_{occ} the constructive occurrence multiplicity.

The point of Definition 6.1 is that these four quantities are *typed*: they live in different codomains, are computed by different readings, and may take different values for one process.

6.2 The cube counter-calibration

The square of §5 is too symmetric to make the point, because over $\mathbb{R}$ it already has a genuine branch. The cube is decisive: it is algebraically of degree three, it is a real homeomorphism, and it has three constructive occurrences.

Calibration 6.2 (Power maps separate the invariants)**.** For $P_{\mathbb{R}}$ the real cube $x \mapsto x^3$ read through the contract $Q_{\mathbb{R}}$, and for $P_{\mathbb{C}}$ the complex cube read through $Q_{\mathbb{C}}$ whose scalar domain is $\mathbb{C}$,

$$\mathbf{d}_{Q_{\mathbb{R}}}(P_{\mathbb{R}}) = (3, 1, 1, 3), \qquad \mathbf{d}_{Q_{\mathbb{C}}}(P_{\mathbb{C}}) = (3, 3, 3, 3).$$

The total multiplicity 3 appears in both profiles but in different coordinates, so the two cannot be reconciled by a single number.

Warning 6.3 (Criticality does not imply branching). $C(x) = x^3$ has $C'(0) = 0$, so the map is critically singular at the origin, yet it is a real homeomorphism: crucially, criticality does not split a real open neighbourhood into several branches. Consequently no argument of the form "the derivative vanishes, therefore there are several sheets over the image" is admissible. A branch count is a statement about the covering over the declared scalar domain and must be computed there.

Remark 6.4. Warning 6.3 is the exact obstruction to the naive reading of §4: the complex structure J detects the presence of a second direction, not the multiplicity of preimages. The two data are independent.

6.3 Invariance is the trivial case of equivariance

A characteristic need not be invariant under a symmetry of its carrier; it may transform.

Proposition 6.5 (Equivariance law). *Let $\sigma(x) = -x$ and let $P_n(x) = x^n$. Then*

$$P_n(\sigma x) = \sigma^n P_n(x), \qquad \textit{i.e.} \qquad P_n(-x) = (-1)^n P_n(x).$$

Equivalently, P_n is equivariant for the sign representation of C_2 when n is odd, and invariant — the trivial representation — when n is even.

Proof. $(-x)^n = (-1)^n x^n$. □

Proposal 6.6 (Equivariant characteristics). The correct general notion of "a characteristic that a symmetry preserves" is equivariance $f(gx) = \pi(g)f(x)$ for a declared representation π; invariance $f(gx) = f(x)$ is the case π trivial. Characteristics should be recorded together with π.

Remark 6.7. Proposition 6.5 is elementary, and it is included because the distinction it names is the one that fails in practice: characters are frequently assumed invariant, and the resulting theory then appears to lose odd-degree data. The data are not lost; the variance was mis-declared.

6.4 The binomial coaction

The final piece explains the obstruction of Theorem 5.13 in structural terms.

Proposition 6.8 (Binomial coaction). *On $\mathbb{Q}[x]$ define $\Delta(x) = x \otimes 1 + 1 \otimes x$, extended as an algebra map. Then for every b,*

$$(\mathrm{id} \otimes \mathrm{ev}_b)\Delta(x^n) = (x + b)^n,$$

where ev_b evaluates the right tensor factor at b.

Proof. $\Delta(x^n) = (x \otimes 1 + 1 \otimes x)^n = \sum_k \binom{n}{k} x^{n-k} \otimes x^k$ by the binomial theorem in the commutative algebra $\mathbb{Q}[x] \otimes \mathbb{Q}[x]$. Applying $\mathrm{id} \otimes \mathrm{ev}_b$ gives $\sum_k \binom{n}{k} b^k x^{n-k} = (x + b)^n$. □

Corollary 6.9 (Why translation expands a word). *The passage of a translation through the n-th power is governed by a coaction with $n + 1$ terms; hence it is a multi-term expansion for $n \geq 2$ and cannot be a single post-composite. This is the structural reason behind Theorem 5.13.*

Proof. The expansion has $\binom{n}{k}$ nonvanishing summands for $k = 0, \ldots, n$; for $n \geq 2$ there are at least three terms with distinct powers of x, whereas a post-composite of x^n with an affine map produces exactly one term of degree n and one constant. □

Remark 6.10 (Vocabulary closed under translation). Proposition 6.8 also says which vocabularies are stable under translation: a power vocabulary is translation-closed exactly when it is closed under the coaction, which for a finite generating set forces the vocabulary to be translation-invariant as a set. §17 records the corresponding open problem for finite vocabularies.

6.5 Consequences carried forward

1. A characteristic is typed by the reading that produces it, and its multiplicity coordinates are not interchangeable (Calibration 6.2, Warning 6.3).
2. Stability under a symmetry is equivariance for a declared representation, not invariance (Proposition 6.5, Proposal 6.6).
3. Translation cannot cross a power inside a word normal form, for a structural reason given by a coaction (Proposition 6.8, Corollary 6.9). Combined with Proposition 5.15, this is the paper's justification for replacing words by typed graphs with explicit copy.

7 Classical calibrations: isogenies and the Legendre crossing

A programme that claims a genuinely new triadic structure must be tested against the places where mathematics already has one. This section reports two calibrations, and the first is not flattering to the programme: a complex elliptic isogeny already carries three exactly-bridged representations, so the bare existence of a bridge is prior art. What survives as novel is narrower and is stated precisely. The second calibration is constructive: in the Legendre family one cusp really does support three readings at once, and the residual that obstructs their identification is exact.

7.1 The isogeny calibration: the bridge is not new

Theorem 7.1 (Three readings of an isogeny)**.** *Let $\phi\colon E_1 \to E_2$ be a complex elliptic isogeny with kernel $\ker\phi$, and let M_α be the matrix of the induced map on the period lattice attached to a complex multiplication α. Then*

$$N(\alpha) = |\det M_\alpha| = \deg\phi = |\ker\phi|,$$

so the norm, the lattice determinant, the covering degree, and the kernel cardinality are four descriptions of one number, computed in three different domains.

Proof. For a complex isogeny the degree equals the size of the kernel and the index of the image lattice, and the lattice index is the absolute determinant of the matrix of the map in a lattice basis. For a CM endomorphism the degree is the algebraic norm. □

Calibration 7.2 (The novelty boundary)**.** Theorem 7.1 shows that a triadic bridge of the kind developed in §5 is already realised classically in the theory of complex elliptic isogenies. The programme's claimable novelty is therefore *not* the existence of three bridged representations. It is, at most:

1. observer-relativity: the bridge is stated relative to a declared finite contract Q and a hypothesis class;
2. finite inference: the reconstruction is an intersection of fibres subject to a budget, with a four-valued outcome;
3. accountable residual: where the classical equalities hold, the residual is provably empty, and where they fail, the residual is characterized.

Warning 7.3 (Two traps in the classical case). Two errors are easy to make and both are avoided in the statement above. First, coordinate branching of a composite $x \circ \phi$ is *not* ramification of ϕ; the two are different structures attached to different maps. Second, the equalities of Theorem 7.1 are equalities of *numbers* obtained from three carriers; they do not identify the carriers, and by

Definition 2.4 an identification of carriers would require a residual argument that the theorem does not supply.

Proposal 7.4 (Characteristics as a groupoid)**.** A characteristic is better modelled as an object of a representation groupoid — a carrier together with declared readings and the isomorphisms between them — than as a single normal-form value. Under this reading, the value of a characteristic is the isomorphism class, and the residual is the stabilizer datum.

7.2 The marked rank-two carrier and its three cusps

The geometry that the triadic programme wants is supplied by the level-two modular curve.

Proposition 7.5 (Three cusps are three slope classes)**.** *The level-two modular curve*

$$Y(2) = \Gamma(2)\backslash\mathbb{H} \cong \mathbb{P}^1 \setminus \{0, 1, \infty\}$$

has exactly three cusps, corresponding to the three points of $\mathbb{P}^1(\mathbb{F}_2)$*, that is, to the three slope classes* $0/1$*,* $1/1$*,* $1/0$*. Equivalently, they are the three oriented slope classes of one marked rank-two carrier.*

Status. The isomorphism is the classical description of $Y(2)$ as the Legendre λ-line; the three cusps are $\lambda \in \{0, 1, \infty\}$. The slope-class description is obtained by reading (ω_1, ω_2) modulo $\Gamma(2)$ and reducing the ratio modulo 2. The corresponding mod-two relations are $\delta_T = \delta_S + \delta_C$ and its cyclic variants. □

Remark 7.6 (Mod-two erases chirality)*.* The reduction above is mod-two, and §4 showed exactly what mod-two forgets: the sign of the orientation word. Consequently the identification of Proposition 7.5 describes the three readings up to a sign that the mod-two picture cannot see, and an integer lift is required. The following construction supplies it.

Construction 7.7 (Oriented integer lift)**.** Let $\delta_T = a$, $\delta_S = b$, $\delta_C = -a - b$ in a rank-two lattice with symplectic form $\langle \cdot, \cdot \rangle$, chosen so that all three cyclic pairings equal $+1$:

$$\langle \delta_T, \delta_S \rangle = \langle \delta_S, \delta_C \rangle = \langle \delta_C, \delta_T \rangle = +1.$$

The order-three chart rotation is

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad R^3 = I.$$

Lemma 7.8 (Tri-chart coherence)**.** *The three coordinate charts* (x, y)*,* $(y - x, -x)$*,* $(-y, x - y)$ *are permuted by* R*, and* $R^3 = I$*.*

Proof. Applying R to a pair yields the next chart, by direct substitution. The characteristic polynomial of R is $\lambda^2 + \lambda + 1$, so $R^3 = I$ by Cayley–Hamilton. □

Remark 7.9*.* Construction 7.7 and Lemma 7.8 are the smallest exact repair of Remark 7.6: the sign lost mod two is retained by working in an oriented integer lattice, at the cost of a nontrivial chart rotation that must be tracked.

7.3 The Legendre crossing

Proposition 7.10 (One cusp, three readings)**.** *Consider the Legendre family*

$$E_\lambda : \ y^2 = x(x-1)(x-\lambda), \qquad \lambda \in \mathbb{P}^1 \setminus \{0, 1, \infty\},$$

with its nonzero two-torsion satisfying $P_0 + P_1 + P_\lambda = 0$. At each cusp $\lambda \in \{0, 1, \infty\}$, the same local singularity is simultaneously

1. *a collision of branch factors of the defining equation;*
2. *a vanishing class δ with Dehn transvection $U_\delta = L_\delta^2$;*
3. *a monodromy of the Picard–Fuchs system*

$$\lambda(1-\lambda)\Pi'' + (1-2\lambda)\Pi' - \tfrac{1}{4}\Pi = 0.$$

Status. Items (i)–(iii) are classical local descriptions of the Legendre degeneration. The content claimed here is only that they apply at one point to one object; the identification of the three *descriptions* with the three *readings* of §5 is not proved and remains the open problem of §17. □

No-go 7.11 (Bare monodromy is spectrally empty)**.** The cusp monodromy acts by

$$M_\delta(v) = v + \langle v, \delta \rangle \, \delta,$$

which is unipotent. Hence a bare cusp monodromy supplies no rich spectrum, and any useful spectral invariant must be attached to paths, transports, correspondences, or gluing data rather than to a single local monodromy operator.

Proof. $M_\delta - \mathrm{id}$ has image contained in the line spanned by δ and has square zero, so every eigenvalue of M_δ equals 1. □

Proposition 7.12 (Central sign and projective closure)**.** *With positive unipotent cusp lifts U_0, U_1, U_∞ one has*

$$U_0 U_1 U_\infty = -I \quad \textit{in the linear lift, while} \quad M_0 M_1 M_\infty = I \quad \textit{after typing the sign into the infinity chart.}$$

Thus the three local operators close projectively, and a central -1 survives in any selected linear lift.

Status. The projective closure is classical. The central sign is the same phenomenon as Proposition 3.9: projectivization forgets a scalar, and a lift retains it. □

Proposition 7.13 (Label transport, not universal swap)**.** *The general law governing how a vanishing cycle moves when the base chart is changed is*

$$L_{g\delta} = g \, L_\delta \, g^{-1},$$

a conjugation of labels. There is no universal swap law for δ across charts.

Proof. Conjugation transports the cycle δ to $g\delta$, and the transvection is determined by its vanishing cycle, so the conjugated operator is the transvection of the transported cycle. □

7.4 The crossing-coherence prism and its decomposable defect

The three cards are now placed on one object, and the result is the paper's principal negative calibration.

Construction 7.14 (Crossing-coherence prism)**.** Take three typed carriers

$$\mathcal{K} = (\text{construction data } q, p, h), \qquad \mathcal{X} = (\text{charge } q \in \mathbb{Z}^2), \qquad \mathcal{T} = (\text{covector } \ell \in (\mathbb{Z}^2)^*),$$

with crossing actions $J^c_{\mathcal{K}}$, $J^c_{\mathcal{X}} = M_c$, $J^c_{\mathcal{T}} = M_c^{-T}$ for each cusp c. The interpretation maps are $F_{\mathcal{K}\mathcal{X}}$, $F_{\mathcal{X}\mathcal{T}} = J$, and $F_{\mathcal{T}\mathcal{K}}$, with residuals carried by the last.

Theorem 7.15 (Crossing squares)**.** *In Construction 7.14:*

1. *the $\mathcal{K}$–$\mathcal{X}$ square commutes strictly on charge;*
2. *the $\mathcal{X}$–$\mathcal{T}$ square commutes strictly, by symplectic duality $JM = M^{-T}J$;*
3. *the $\mathcal{T}$–$\mathcal{K}$ square commutes only up to the exact history residual, recorded as $[c]$.*

Proof. Item (i) is immediate, since both actions act on the same charge coordinate. Item (ii) is the defining property of the symplectic form: for M preserving $\langle\cdot,\cdot\rangle$ the relation $JM = M^{-T}J$ holds. Item (iii) fails by construction, since a period covector does not determine construction history (Proposition 10.17); the failure is recorded rather than removed. □

Proposition 7.16 (Two independent residual channels)**.** *The prism of Construction 7.14 carries two independent residuals:*

1. *a central lift sign, of which $-I$ versus $+I$ is the visible instance;*
2. *a history suffix $[c]$ recording the construction path.*

A coarse projective observer that identifies $v \sim -v$ and $M \sim -M$, and that erases the history, sees neither.

Proof. The first is invisible under projectivization by Proposition 3.9; the second is invisible because it is not a function of the charge or of the covector. Their independence is exhibited by varying one while fixing the other. □

Theorem 7.17 (No-go: the triadic defect decomposes)**.** *The total finite defect of the prism is*

$$D_Q = (-I,\, [\infty, 1, 0]),$$

and it decomposes into the two channels of Proposition 7.16. Consequently no irreducible triadic 3-cocycle is exhibited by this model.

Proof. The central sign is a scalar and the history suffix is a word; the projections onto the two summands are maps of defect data, and each vanishes on the kernel of the other. A defect admitting such a decomposition is by definition not irreducible. □

Calibration 7.18 (Status of the negative calibration)**.** Theorem 7.17 is reported as the paper's principal negative calibration. It establishes that the finite Legendre prism does *not* produce a genuinely triadic coherence defect; it does not establish that no such defect exists.

Proposal 7.19 (What genuine triadicity would require)**.** Genuine triadicity requires a coherence defect Θ_c that is not reconstructible from any pair of the three crossings. In the language of §9 this is the statement that the triple support strictly exceeds what the three pairwise supports determine.

Conjecture 7.20 (Triadic singular crossing)**.** There exists a carrier on which the three readings of §5 are related by crossing characteristics with a non-decomposable cyclic defect, functorial under refinement of the observer contract Q.

7.5 What this section settles

1. The bare triadic bridge is prior art (Theorem 7.1, Calibration 7.2); the programme's novelty must be sought in observer-relativity and residual accounting.
2. The classical carrier for three readings is the level-two modular curve with its three cusps, and the mod-two picture requires an oriented integer lift to retain chirality (Proposition 7.5, Construction 7.7, Lemma 7.8).
3. Crossing coherence is strict for two pairs and residual-laden for the third, and the resulting defect decomposes (Theorem 7.15, Proposition 7.16, Theorem 7.17).
4. Whether a non-decomposable defect exists is open (Proposal 7.19, Conjecture 7.20).

8 Through-crossings, resolution, and monodromy

§7 produced local monodromy operators and then observed (No-go 7.11) that a single one of them carries no spectrum. This section supplies the missing structure. Its central claim is a change of genre: passing through a singular fibre is not an arrow but a cospan together with a choice, and once that is made explicit, monodromy becomes a *difference* of the two choices rather than a primitive. The change pays immediately: the Legendre monodromy is derived from a quadratic base change, and a primitive transvection is realised on a finite cell complex.

8.1 Crossing is a cospan with a resolution

Definition 8.1 (Through-crossing)**.** Let F_0 be a singular fibre and $F_\pm$ the two smooth fibres on either side of the critical value. A *through-crossing* is the cospan

$$F_- \xrightarrow{\pi_-} F_0 \xleftarrow{\pi_+} F_+,$$

whose *through relation* is the fibre product

$$R_c = F_- \times_{F_0} F_+.$$

Proposal 8.2 (A crossing is not a transport arrow)**.** There is no canonical smooth transport arrow $F_- \to F_+$ across a singular fibre. What exists is the through relation R_c together with a further choice, and the choice is the mathematical content.

Definition 8.3 (Resolution torsor)**.** A *resolution* of the through-crossing is a selection, for each point of the critical fibre, of which smooth point on the negative side is joined to which on the positive side. The set of resolutions is a torsor under the monodromy group generated by the vanishing transvection:

$$\mathrm{Res}(R_c) \text{ is a torsor under } \langle T_\delta \rangle \cong \mathbb{Z}.$$

Remark 8.4 (Why the torsor language is not decoration). A resolution has no privileged origin: any two resolutions differ by a power of the transvection, and no resolution is singled out by the geometry of the cospan alone. This is the precise sense in which §7's monodromy operators are not intrinsic to a single fibre.

Remark 8.5 (A worked affine instance, imported from Paper 0). The resolution torsor of Definition 8.3 is an abstract statement until an instance is produced. Paper 0 supplies one in the affine sector, and we cite it rather than re-derive it. There, $\mathrm{Aff}^+(1,\mathbb{R})$ acts on $\mathbb{H}^2$; a dilation is hyperbolic and its

conjugacy class is determined by its multiplier, so its translation length is a function of the class; a nontrivial translation is parabolic, and by the two-layer proposition of Paper 0 [3] every nontrivial translation of a fixed sign is conjugate to every other, the stabiliser being the translation subgroup. The class of a translation therefore carries no continuous invariant, and the set of its conjugates is a torsor under the dilations with no privileged origin — the same shape as Definition 8.3. Two consequences are worth stating, because they are the reason this paper does not identify the two torsors. First, the affine instance is a torsor under $\mathbb{R}_{>0}$, not under $\mathbb{Z}$, so it does not yet supply the $\langle T_\delta \rangle \cong \mathbb{Z}$ structure that Theorem 8.6 uses. Second, the passage from the affine sector to the projective one needs the trace trichotomy $\mathrm{tr}^2 - 4\det$, which Paper 0 does not develop and this paper does not develop either. So the instance is a calibration of the *shape*, not an identification of the objects.

8.2 Monodromy as a difference of resolutions

Theorem 8.6 (Monodromy from two through-resolutions)**.** *Let $P^{\uparrow}$ and $P^{\downarrow}$ be the two resolutions of Definition 8.3 obtained by passing through the critical value along the two admissible sides. Then the around-the-critical-value monodromy is*

$$\left(P^{\downarrow}\right)^{-1} P^{\uparrow} = T_\delta,$$

the vanishing transvection.

Proof. Both resolutions identify the same two smooth fibres, and their composite is the deck transformation of the passage around the critical value. For a single node that deck transformation is the Picard–Lefschetz transvection of the vanishing cycle δ, by the local theory of nondegenerate critical points. □

Remark 8.7. Theorem 8.6 is the section's organising result. It replaces "insert the monodromy matrix" by "compare two resolutions", and it explains why No-go 7.11 had to be true: one resolution alone determines nothing, because the torsor has no origin.

8.3 The Legendre cusp as a quadratic base change

Proposition 8.8 (Local reduction)**.** *The local equation of the Legendre family at a cusp reduces, after an analytic change of coordinates, to*

$$UV = \lambda^2.$$

The map $\lambda \mapsto s = \lambda^2$ is a quadratic base change of the node $UV = s$, and consequently

$$M_{\text{Legendre}} = T_\delta^2.$$

Proof. The Legendre family has an A_1 node at $\lambda = 0$ (and, after the standard change of chart, at the other two cusps as well). Pulling a node back along $\lambda \mapsto \lambda^2$ doubles the local monodromy, since a full turn in s is covered by half a turn in λ. By Theorem 8.6 the node monodromy is T_δ, so the pullback has monodromy T_δ^2. □

Remark 8.9 (What has been gained)*.* Proposition 8.8 derives a monodromy matrix that §7 could only state. The derivation uses the *shape* of the base change, not a computation with periods. This is the sense in which the through form is a better carrier than the operator: the operator is its shadow.

8.4 Cellular realisation on a finite annulus

The abstract torsor has an exact finite model.

Construction 8.10 (Cubical annulus and shear)**.** Let A_n be the cubical annulus obtained from a strip of n squares by identifying the two boundary circles with a shift, so that

$$\chi(A_n) = 0, \qquad (b_0, b_1, b_2)(A_n) = (1, 1, 0).$$

Let D be the cellular shear that displaces each vertex one step along the annulus, and let $g\colon A_n \to T_n$ be the gluing to the torus obtained by identifying the two boundaries.

Proposition 8.11 (Shear induces the transvection)**.** *The induced map on homology is the Picard–Lefschetz transvection*

$$D_*(x) = x + \langle \delta, x\rangle\, \delta, \qquad M_D = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad M_D^T J M_D = J,$$

where J is the symplectic form. Hence the shear is symplectic and realises the primitive Dehn twist.

Proof. The shear displaces a representative of δ once around the annulus while fixing the transverse class, which is precisely the defining formula of the transvection. The matrix is read off on the basis (δ, γ) with $\langle \delta, \gamma\rangle = 1$, and symplecticity follows from $\langle x + \langle \delta, x\rangle\delta,\ y + \langle \delta, y\rangle\delta\rangle = \langle x, y\rangle$ by antisymmetry and $\langle \delta, \delta\rangle = 0$. □

Theorem 8.12 (No-go: vertex closure does not imply path closure)**.** *In Construction 8.10 the vertex-level shear satisfies $D_0^n = I$, while the lifted one-chain level satisfies $D_1^n \neq I$ and $M_D^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \neq I$ for $n \neq 0$. Thus closure of a finite vertex permutation is not the closure of the lifted path.*

Proof. The vertex permutation is a cyclic shift of a finite set, hence has finite order dividing n. The induced homology map is unipotent with nontrivial nilpotent part, so it has infinite order. □

Remark 8.13 (The pair of separations). Theorem 8.12 is the finite model of Proposition 8.2: a closed finite quotient can look trivial while the lifted object is not. §10 records this as one instance of a general pattern rather than as an isolated curiosity.

Proposition 8.14 (Marked identity and the nodal pinch)**.** *Let $q\colon T_n \to$ (the nodal quotient) be the pinch. Then $q_* D_* = q_*$, so the shear is invisible to the nodal reading; but the marked outgoing condition gives a strict identity $q_+ D = q_-$, which upgrades D to a genuine through-resolution.*

Proof. The pinch kills precisely the vanishing direction, and the shear acts on the transverse direction by the transvection, which is trivial modulo δ. The marked statement tracks which boundary the identity is asserted on, whence the strict identity. □

8.5 Through forms as fibre products over a singular middle

Definition 8.15 (Through form)**.** For a typed singular middle object N and two fibres $F_\pm$ over it, the *through form* is

$$\langle L \mid_N R\rangle := F_- \times_N F_+.$$

Proposition 8.16 (Order and compatibility are independent)**.** *Conway number-admissibility of a cut $\{L \mid R\}$ and through-compatibility of $\langle L \mid_N R\rangle$ are logically independent: either may hold while the other fails.*

Proof. Admissibility is a condition on the order structure of L and R alone; compatibility is a condition on the projections to N. Choosing N degenerate makes compatibility fail with admissible sides, and choosing an incompatible order on N makes compatibility hold with inadmissible sides. □

Proposition 8.17 (Resolution fibre count)**.** *If the critical fibre consists of r regular points together with n phase points, the through form carries $r + n^2$ resolutions, indexed by the relative offset $P_k \colon i \mapsto i + k$ modulo n.*

Proof. Each regular point has one resolution; each pair of phase points may be joined, and the join is determined by a relative offset k modulo n, giving n classes for each of n points. □

Remark 8.18 (Duality is relational converse)*.* The duality appropriate to through forms is not an inverse but the relational converse $T^\star = T^{\mathrm{op}}$ with contravariant composition. This is forced by Proposition 5.10: a transport that is relation-valued cannot have a functional dual.

8.6 One feature, three readings

Theorem 8.19 (Figure-eight calibration)**.** *There is a specific figure-eight braid word w for which*

$$-D_w = -\phi\left(\frac{\partial w}{\partial b}\right) = \det(tI - M) = t^2 - 3t + 1,$$

where D_w is the AEG affine defect, ϕ is the abelianised Fox derivative [2], and M the fibre monodromy. The common value is a feature of the ordered history and does not reconstruct the knot.

Status. An exact computation for one word, with all three quantities evaluated independently. The equality is a calibration on a single word, not a general identity, and the non-reconstruction clause is essential: the polynomial is a characteristic in the sense of Definition 2.4, with the word as its residual. □

Proposition 8.20 (Double periods)**.** *Two visible periods p, q give a phase torus $\mathbb{Z}/p \times \mathbb{Z}/q$ under the successor $\tau(i, j) = (i + 1, j + 1)$, with*

$$\#\textit{orbits} = \gcd(p, q), \qquad \textit{orbit length} = \operatorname{lcm}(p, q), \qquad \textit{primitive winding} = \big(q/g,\ p/g\big), \quad g = \gcd(p, q).$$

The unit-tangent lift supplies the third direction, and the orbifold meridians

$$m_P = (p - 1, -1), \qquad m_Q = (-1, q - 1), \qquad \det = pq - p - q$$

recover the arithmetic. Equal numerals across the three types are not identified.

Proof. An orbit of τ has length $\operatorname{lcm}(p, q)$ because both coordinates must return, and the number of orbits is $pq/\operatorname{lcm}(p, q) = \gcd(p, q)$. The winding is read off by dividing the periods by g. The determinant computation is direct. □

8.7 Return is not period

No-go 8.21 (No intrinsic periodic dynamics)**.** There is a finite diagram containing the source-free components constant; discard and constant; neg; discard whose least boundary-return lengths are 2 and 3 respectively; but monotonicity of the completed-past rank forces the verdict *NotRepresentable* for intrinsic periodic dynamics. Hence a boundary-return word is not a periodic orbit.

Proof. The completed past of a state grows monotonically along a checked execution, so no state can recur. The return lengths are therefore properties of words, not periods of a dynamical system; a genuine period requires a feedback operator that the syntax does not supply. □

Remark 8.22. No-go 8.21 is the most frequently violated boundary in this material. It is why Part II treats iteration as a syntactic structure (§17) rather than as a semantic one.

8.8 Summary

1. A crossing is a cospan with a resolution torsor (Definition 8.1, Proposal 8.2, Definition 8.3).
2. Monodromy is the difference of two resolutions, and the Legendre monodromy follows from the quadratic base change (Theorem 8.6, Proposition 8.8).
3. A primitive transvection is realised by an exact cellular shear, and vertex closure does not imply path closure (Proposition 8.11, Theorem 8.12).
4. Through forms are fibre products over a singular middle, whose duality is relational converse (Definition 8.15, Propositions 8.16, 8.17).
5. One feature can be read three ways on one word, and a boundary-return word is still not a period (Theorem 8.19, Proposition 8.20, No-go 8.21).

9 A logic generated by a satisfaction form

Sections 5–8 produced carriers and crossings. This section asks what kind of *logic* such a carrier supports, and the answer is constrained from the start: evidence is positive, propositions are sections rather than subsets, entailment is refinement, and negations are quotients that must be declared. The section closes by adding a quantitative enrichment and by exposing the trap that makes it vacuous if used naively.

9.1 Positive evidence

Definition 9.1 (Satisfaction form)**.** Let $\mathcal{T}_Q, \mathcal{X}_Q, \mathcal{K}_Q$ be the three typed carriers of a contract Q. A *satisfaction form* is a relation

$$\Lambda_Q \subseteq \mathcal{T}_Q \times \mathcal{X}_Q \times \mathcal{K}_Q$$

read as a map into the Sierpinski lattice $\mathbf{2} = \{\bot, \top\}$ with $\bot < \top$, where $\top$ means "positive evidence of satisfaction" and $\bot$ means "no positive evidence".

Warning 9.2 ($\bot$ is not false). The value $\bot$ in Definition 9.1 records the absence of positive evidence. It does not assert the negation of the proposition. This is the same discipline as Warning 2.13, and it is the reason the constructions below never form a Boolean complement without declaring one.

9.2 Sections are the atoms

Definition 9.3 (Opposite-pair sections)**.** For each pair of coordinates, define

$$A^{t}_{x,k} = \{\, t \in \mathcal{T}_Q : \Lambda_Q(t,x,k) = \top \,\}, \quad A^{X}_{t,k} = \{\, x : \Lambda_Q(t,x,k) = \top \,\}, \quad A^{\mathcal{K}}_{t,x} = \{\, k : \Lambda_Q(t,x,k) = \top \,\}.$$

Proposal 9.4 (Generated propositions)**.** The admissible propositions of the logic of Q are generated by the sections of Definition 9.3, not by the ambient powerset:

$$\mathcal{L}_Q = \langle G_Q \rangle, \qquad \text{and in general} \qquad \mathcal{L}_Q \neq \mathcal{P}(\mathcal{T}_Q \times \mathcal{X}_Q \times \mathcal{K}_Q).$$

Remark 9.5 (Why this is the central design choice). Taking the powerset as the proposition set is the default of classical semantics and it is wrong here for a concrete reason: a subset of the carrier may be undefinable by any section, so admitting it as a proposition silently assumes an expressive power that the carrier does not have. Proposal 9.4 refuses that assumption and makes the expressive reach of the carrier an explicit object.

9.3 Entailment as refinement

Definition 9.6 (Support and preorder)**.** Let $\operatorname{supp} P$ be the set of triples at which a generated proposition P carries value $\top$. Define

$$P \preceq_Q R \iff \operatorname{supp} P \subseteq \operatorname{supp} R.$$

Proposition 9.7 (Refinement is an entailment preorder)**.** *$(\mathcal{L}_Q, \preceq_Q)$ is a preorder, and joins exist for the opposite pairs: writing P_I and P_{-I} for the two lifted signs of a central-sign pair,*

$$P_{\text{proj}} = P_I \vee P_{-I}$$

is exactly the proposition seen by the coarse projective observer.

Proof. Reflexivity and transitivity follow from set inclusion of supports. The join statement is Definition 9.6 applied to the union of the two supports, which is precisely the set on which the projective reading forgets the sign. □

Calibration 9.8 (Observer quotient with residual)**.** On the finite carrier of this section, a three-coordinate observer quotient of the section lattice is computed together with its exact preimage table. The residual of the quotient is therefore known and finite, as Proposition 2.6 requires.

Proposal 9.9 (Adjoint quantifiers)**.** The quantifiers appropriate to this logic are adjoints to pullback along an observer map q:

$$\exists_q \dashv q^* \dashv \forall_q,$$

where q^* is the observer quotient of Calibration 9.8.

Remark 9.10. Proposal 9.9 is a Proposal: the adjunction is the right shape for observer transport, but no theorem here establishes that the finite lattice of this section is closed under it. What is established is the data on which such a theorem would be stated.

9.4 The triadicity test

Proposal 9.11 (Falsifiable triadicity)**.** Triadicity of Λ_Q is falsifiable by ambiguous global fillings: if two distinct global fillings have equal pairwise sections but different triple support, then the triple carries information that no pair determines, and the form is triadic in the sense of Proposal 7.19. If every two fillings with equal pairwise sections have equal triple support, the form is only pairwise in disguise.

Calibration 9.12 (The finite fixture fails the test)**.** On the finite satisfaction form constructed in this section the test of Proposal 9.11 *fails*: the triple support is determined by the pairwise sections. This is the same phenomenon as Theorem 7.17 in a logical setting, and it is the reason Conjecture 7.20 is stated as a conjecture.

9.5 Logic as a learned characteristic

The conceptual reading of the section is worth stating separately, because it organises the three notions that the constructions above keep apart.

Proposal 9.13 (Three grades of logicality)**.** 1. A *proposition* is a positive (Sierpinski-valued) characteristic.

2. *Entailment* is the first relational characteristic: it judges pairs.

3. *Abstraction* is a coarse invariant together with a declared forgotten fibre $R_Q(b)$; a logic earns stability exactly by declaring what it forgets.

Warning 9.14 (Three different things)*.* Bivalence, semantic completeness, and decidability are independent. A logic may be two-valued and incomplete, or complete and undecidable, and the passage from positive evidence to a Boolean one is a *second, declarable quotient* rather than a consequence of the first. Nothing in this section licenses forming a complement without that second declaration.

9.6 Energy as an enrichment, and its trap

A satisfaction form may be enriched by a quantitative measure of how a characteristic varies along declared directions.

Definition 9.15 (Directional energy)**.** On a finite observation carrier with declared move directions D, the *directional Dirichlet energy* of a function f is

$$E_D(f) = \sum_{(u,v)\in D} \big(f(u) - f(v)\big)^2,$$

and the *energy vector* is $E_Q(f) = \big(E_t(f), E_X(f), E_{\mathcal{K}}(f)\big)$.

Proposition 9.16 (Exact coarse–residual split)**.** *Let q be an observer that averages along a forgotten fibre, and write $\bar{f}$ for the coarse average and $r = f - q^*\bar{f}$ for the zero-mean residual of each fibre. Then*

$$E_Q(f) = E_Q\big(q^*\bar{f}\big) + E_Q(r),$$

and for the constructional direction in which the fibre is forgotten, $E_{\mathcal{K}}(q^*\bar{f}) = 0$.

Proof. Each summand $\big(f(u) - f(v)\big)^2$ decomposes as the sum of the squares of the coarse difference and the residual difference plus a cross term that vanishes because r has zero mean on each fibre and the fibres are disjoint. The second claim is immediate since $q^*\bar{f}$ is constant along the forgotten direction. □

Proposition 9.17 (Energy detects invariance)**.** *For a binary characteristic χ_U,*

$$E_D(\chi_U) = 0 \iff \chi_U \text{ is constant along every } D\text{-move},$$

and in that case E_D equals the size of the boundary cut of U along D.

Proof. Each summand is $(f(u) - f(v))^2 \in \{0, 1\}$ for binary f, so the sum vanishes exactly when no edge crosses the boundary of U; otherwise each crossing edge contributes 1. □

No-go 9.18 (The zero-energy trap)**.** The constant characteristic minimizes every directional energy. Consequently any optimisation principle of the form "minimise E_Q" is vacuous unless accompanied by an explicit nontriviality constraint, and no learning rule may be justified by energy decrease alone.

Proof. By Proposition 9.17 a constant characteristic has zero energy, which is the global minimum since every summand is a square. □

Remark 9.19. No-go 9.18 is worth stating at length because the error it forbids is attractive: the energy splits exactly, it is computable, and its minimisers are therefore easy to find — too easy. The exactness of Proposition 9.16 and the vacuity of No-go 9.18 are two sides of the same computation.

9.7 Summary

1. Evidence is positive and $\perp$ is not false (Definition 9.1, Warning 9.2).
2. Propositions are generated by sections, not by the powerset, and entailment is refinement of support (Proposal 9.4, Proposition 9.7).
3. The triadicity test is falsifiable, and the finite fixture of this section fails it (Proposal 9.11, Calibration 9.12).
4. Energy enriches the picture exactly, and is vacuous without a nontriviality constraint (Propositions 9.16, 9.17, No-go 9.18).

10 Non-faithfulness: the residual catalogue

The previous sections produced no-go results one at a time. This section collects them, because together they say something stronger than each does separately: there is a *systematic* pattern in which quantity fails to determine construction, and naming the pattern is more useful than proving its instances. The section also records the positive counterweights, since a catalogue of failures without a statement of what a faithful carrier looks like would be misleading.

10.1 The pattern

Proposal 10.1 (The residual pattern)**.** Let σ be a quantity computed from a process by an observer Q. In every instance catalogued below, σ is a *partial quotient*: it is invariant under a group of construction changes that the process itself does not admit. The correct response is never to replace σ by a finer scalar; it is to retain a residual recording the orbit of the forgotten changes.

Remark 10.2. Proposition 10.1 is a Proposal because "partial quotient" is a description, not a hypothesis with a theorem attached. Its content is methodological and its justification is the list: in each case below, the residual is exactly what repairs the deficiency, and no case is repaired by refining the scalar.

10.2 Spectral shadows

Theorem 10.3 (Spectrum does not determine history)**.** *Let x be a single input. The programs* $\mathrm{add}(\mathrm{copy}(x))$ *and* $\mathrm{scale}_2(x)$ *agree on realized value, on derivative, on all iterates, and on the action spectrum, which is* $\{2\}$ *in both cases; they differ in the construction graph, in the occurrence structure, and in the source history. Hence no observer that sees only values, derivatives, iterates, and spectra can separate them.*

Proof. Both programs induce the same endomorphism $x \mapsto 2x$ on the declared state, so every functional of the induced action agrees. They differ in the number of occurrences of the source and in the recorded lineage, which are not functions of the induced action. □

Proposition 10.4 (The affine carrier is not diagonalisable into a spectrum)**.** *In the affine carrier, scaling acts diagonally with spectrum* $\{m^{-1}, \dots, m^{-K}\}$*, while addition acts unipotently and is nilpotent on the relevant graded pieces. Consequently a bare eigenvalue set cannot encode addition: the characteristic polynomial of a unipotent operator is* $(\lambda - 1)^d$ *regardless of the translation amount.*

Proof. Scaling conjugates $A \mapsto MAM^{-1}$ and stretches the translation grid by m at each graded level, giving the stated spectrum. Addition is the exponential of a nilpotent translation, whose characteristic polynomial carries no information about the amount. □

10.3 Two faithful carriers, for contrast

Theorem 10.5 (Marked second-jet faithfulness)**.** *The positive affine fragment embeds faithfully into the marked second jet at the projective point at infinity,*

$$m = c_1^{-1}, \qquad a = -c_2 c_1^{-2},$$

where (c_1, c_2) *are the marked jet coefficients. The faithfulness is a statement about* marked *jets: the unmarked jet does not suffice, since it is defined only up to the chart that the marking fixes.*

Proof. Composition of affine maps corresponds to multiplication in $\mathbb{R}[[q]]/(q^3)$ on the marked coefficient pair; solving for (m, a) gives the displayed formulas, and the correspondence is a bijection onto the positive affine fragment within the marked class. □

Theorem 10.6 (Co-generation of polynomial and matrix readings)**.** *Transport is homogeneous of degree one in declared symbolic probe generators, and the extracted coefficient table equals the transpose of the forward Jacobian. Hence the polynomial-like and matrix-like presentations of one process are co-generated, rather than independently imposed.*

Status. Proved for the declared probe class by linearity of the demand section. The phrase "after a chart" is essential: the identification of the coefficient table with the transpose requires the port chart to be declared, and without it the two readings are merely isomorphic. □

Remark 10.7 (The moral of the contrast). Theorems 10.5 and 10.6 are the positive half of the catalogue. Both work by making the marking or the chart explicit. The failures below are exactly the cases where a marking or chart was silently omitted, and the residual records what the omission cost.

10.4 The observation tower is strict

Theorem 10.8 (Strict observation tower)**.** *The three observation levels*

$$\mathcal{O}_{\text{prog}} \supsetneq \mathcal{O}_{\text{or}} \supsetneq \mathcal{O}_{\text{proj}}$$

are strictly separated: projective spectral data is strictly coarser than oriented action data, which is strictly coarser than program-aware data. Each strictness is witnessed by a pair of programs identified at the lower level and separated at the higher.

Proof. The projective level forgets the central sign (Proposition 3.9), and the oriented level forgets occurrence structure (Theorem 10.3). Both forgettings are non-injective on the finitary fixtures exhibited there. □

10.5 Objectification and effective completion

No-go 10.9 (Objectification is not natural)**.** Let Obj send a finite cut presentation to its simplest surreal value. Then Obj is non-injective, and it is not natural:

$$g(\mathrm{Obj}\, C) \neq \mathrm{Obj}(gC) \qquad \text{for } g(z) = 3z \text{ and a suitable } C.$$

Consequently no analysis factoring through the objectified value admits a left-inverse synthesis.

Proof. Non-injectivity is immediate, since distinct presentations may share a simplest value. For non-naturality, the two sides of the displayed inequality are computed on an explicit finite C; the objectified value is simplified before scaling on the left and after scaling on the right, and simplicity is not preserved by scaling. □

Proposition 10.10 (Effective one-sidedness)**.** *For a universal prefix-free machine U, let*

$$\Omega_U = \sum_{U(p)\downarrow} 2^{-|p|} \leq 1$$

be its halting mass, and let $(L_\Omega \mid R_\Omega)$ be the associated cut. Then L_Ω is computably enumerable while R_Ω is not. Hence a computational cut is intrinsically asymmetric.

Status. The Kraft bound is a standard measure-theoretic computation on the cylinder sets of a prefix-free set. The asymmetry is the standard halting asymmetry: membership on the left is witnessed by a halting computation, while membership on the right would decide non-halting. □

Proposition 10.11 (Mass cannot repair the fibre)**.** *The objectification fibre mass*

$$m_{U,Q}(x) = \sum_{\mathrm{Obj}_Q(U(p))=x} 2^{-|p|}$$

is a lower-semicomputable semimeasure. It quantifies the fibre but cannot recover its members.

Proof. Lower semicomputability follows from the definition as a sum over enumerated halting computations. Recovery fails because a sum over a set determines neither its cardinality nor its elements. □

Proposition 10.12 (No total finite coding of **No**)**.** *There is no set-sized family of finite codes surjecting onto the surreal numbers.*

Proof. A set-sized family of finite codes is a set, and **No** is a proper class; no set surjects onto a proper class in the relevant setting. □

10.6 Snapshots, scopes, and traces

Proposition 10.13 (Equal snapshots, nonempty interval)**.** *For exact program slices between nested cuts, equal frontier snapshots do not imply an empty interval: an explicit discard witnesses two distinct slices with the same snapshot.*

Proposition 10.14 (Zero-event incidence is not single-valued)**.** *A zero-event graft frame is compatible with two distinct identity slices fixed by an independent surgery. Hence the available frame and slice data admit no canonical single-valued placement of scope in cut; the fibre has two elements.*

Proposition 10.15 (Trace arithmetic does not identify the sides)**.** *For the three-sided trace calibration, the temporal and spatial residuals vanish while the constructional residue* $(+1, -1, 0, 0, 0)$ *does not. Consequently the two sides are not semantically identified by their trace arithmetic.*

Proposition 10.16 (Count factorization and the recovery obstruction)**.** *Let* T *be the trace count and* n *the construction length. Then*

$$T(p) = (n,\ n-1,\ 3(n-1)),$$

a positive factorization through construction length. However the projection $(T, S) \mapsto C$ *is not single-valued: an explicit pair of constructions shares* (T, S) *and differs in* C*.*

Status. The factorization is an exact finite count. The obstruction is an explicit pair, checked on the recorded data; it is a counterexample to single-valued recovery, not a general classification of the fibres of $(T, S) \mapsto C$. □

Proposition 10.17 (Period covectors do not determine history)**.** *A period covector is a functional of a homology class and therefore constant on the construction paths that realise that class. It cannot reconstruct the construction history.*

10.7 Minimal expressive repair

One result in the catalogue deserves separate billing, because it gives an exact *minimum* rather than only an obstruction.

Theorem 10.18 (Order-embedding obstruction)**.** *Let a coarse observation erase an incomparability, i.e. two elements become indistinguishable in the observation order. Then no single added scalar coordinate can restore the lost order, whereas two monotone Boolean coordinates can, and two is the minimum.*

Proof. A single scalar added to a coarse observation produces a total preorder on the fibre, since scalars are totally ordered; a total preorder cannot contain an incomparability. Two monotone Boolean coordinates separate the four possible cells, which suffices to embed the erased relation, and one coordinate gives only two cells. Hence two is both sufficient and necessary. □

Remark 10.19 (Why this is the catalogue's high point). Theorem 10.18 converts "coarse observation loses a relation" from a complaint into a quantitative statement with an exact constant. It is the model for what the rest of the catalogue would need in order to become equally informative.

10.8 Scale and winding

Proposition 10.20 (Scale forgetting)**.** *Let* $\lambda = \varepsilon L z$ *with* $L = (2I + C)/3$ *and* C *the cyclic permutation matrix,* $C^3 = I$*. Then* L *is invertible over* $\mathbb{Q}$*, with*

$$(2I + C)(4I - 2C + C^2) = 9I,$$

so the scale-marked triple is recoverable provided $\varepsilon \neq 0$*. If the scale marker is discarded, the pairs* (ε, z) *and* $(\varepsilon/2, 2z)$ *share* λ*, and recovery fails.*

Proof. Expanding $(2I + C)(4I - 2C + C^2) = 8I - 4C + 2C^2 + 4C - 2C^2 + C^3 = 9I$ using $C^3 = I$ gives the inverse, hence invertibility. The collision is immediate from $\lambda = (\varepsilon/2)L(2z)$. □

No-go 10.21 (Winding counts do not determine the global action)**.** Abelian winding counts do not determine the global action on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$: there are words of total winding zero whose commutator is nontrivial,

$$ABA^{-1}B^{-1} = \begin{pmatrix} 13 & 8 \\ 8 & 5 \end{pmatrix} \neq I.$$

Moreover endpoint closure does not imply identity of the transported homology: a diamond path of winding $+1$ transports $(0, 1)$ to $(2, 1)$.

Proof. Two computations on explicit words. The first exhibits a nontrivial commutator with zero abelian shadow, so the abelianisation is not faithful. The second exhibits a closed endpoint with nontrivial transport. □

10.9 The catalogue

Shadow	What it fails to determine
value, derivative, iterates, spectrum	construction graph, occurrence, source history (Thm 10.3)
eigenvalue set	the translation amount of a unipotent factor (Prop 10.4)
projective spectral data	the central sign; oriented action (Thm 10.8)
oriented action data	occurrence structure (Thm 10.8)
objectified surreal value	the presentation, and it is not natural (No-go 10.9)
fibre mass of an objectification	the members of the fibre (Prop 10.11)
frontier snapshot	emptiness of an interval (Prop 10.13)
frame and slice data	a single-valued scope placement (Prop 10.14)
trace arithmetic	identification of the two sides (Prop 10.15)
count pair (T, S)	the construction C (Prop 10.16)
period covector	construction history (Prop 10.17)
discarded scale marker	the marked triple (Prop 10.20)
abelian winding	the global action (No-go 10.21)
boundary-return word	a periodic orbit (No-go 8.21)
directional energy	any nontrivial characteristic (No-go 9.18)

10.10 What the catalogue implies

1. The failures are not ad hoc. In each row the shadow is invariant under changes the process does not admit, and the repair is a residual rather than a finer scalar (Proposal 10.1).
2. Where a residual is supplied, the theory works: marked jets and declared port charts give genuine faithfulness (Theorems 10.5, 10.6).
3. At least one obstruction has an exact minimum (Theorem 10.18), which is the standard the rest should meet.
4. Two of the rows are about *absence of structure* rather than loss of information: No-go 8.21 (no feedback operator) and No-go 9.18 (no nontriviality constraint). These are the invitations to Part II, where the missing structures are supplied syntactically.

Part II
Threaded apertures and the six-port kernel

11 Aperture-first semantics: the typed hole

Part I repeatedly met a place where a reading was *undetermined*: a missing feedback operator (No-go 8.21), a fibre whose members a mass could not recover (Proposition 10.11), a logical value $\perp$ that meant no evidence rather than falsehood (Warning 9.2). In each case the honest description of the situation was not a value but a *place where something belongs and is not yet there.*

Part II takes that place as its primitive. This section introduces it, fixes its discipline, and shows what a finite semantics built on it looks like.

11.1 Why the hole is primitive

Proposal 11.1 (Aperture-first)**.** The primitive object of a partial semantics should be the typed hole — a position of declared type whose content is not determined — rather than a state plus an exception, or a value plus a flag. The reasons are that holes compose, that they carry their type, and that the operations which fill and empty them are not forced to be inverses.

Definition 11.2 (Typed aperture)**.** Let D be a finite set of domain slots and let τ assign a type to each slot. A *typed aperture at* $d \in D$ is a syntactic object h_d satisfying:

1. h_d is syntactically empty — it contributes no term of its own;
2. h_d has a declared type $\tau(d)$;
3. semantically, h_d denotes the *filling family* of all admissible inhabitants of $\tau(d)$ compatible with the surrounding context.

The third clause is the substantive one: an aperture is not a wildcard. It is a family, and the family is part of the semantics.

Proposal 11.3 (Open and close are not inverses)**.** The operations that open an aperture and that close it are deliberately not inverse. Closing selects an inhabitant and thereby *records* a choice; opening again does not erase the record but adds a new one, and the difference is exactly the residual of Definition 2.4.

Remark 11.4. Proposal 11.3 is the syntactic form of Proposition 2.14: closing is a quotient, and the record is the residual. Treating open and close as inverses would amount to asserting that the quotient has a canonical section, which §2 showed it does not.

11.2 The four structural statuses

Definition 11.5 (Aperture status)**.** At any stage an aperture is in exactly one of four structural statuses:

1. *no aperture* — the position is not a hole in this context;
2. *inhabited* — the filling family is nonempty and a witness is available;
3. *exhaustively empty* — the filling family is empty *and* the emptiness is certified by an exhaustion argument;

4. *unresolved* — neither a witness nor an exhaustion is available within the declared budget.

Warning 11.6 (The third and fourth statuses are different). Status (3) and status (4) are the aperture-level form of Warning 2.13, and conflating them is the most common error in this material. An exhausted search that finds no inhabitant is a positive result; a search that runs out of budget is not a result at all.

11.3 The finite terminal-world carrier

The semantics of Part II is finite from the outset. Its carrier is built from the possible outcomes of a bounded process.

Definition 11.7 (Terminal-world carrier)**.** Let D be a finite set of domain slots. A *terminal world* is a complete assignment of values to D. The *terminal-world carrier* H_D is the set of terminal worlds that are *nonempty*, meaning that the assignment is realised by at least one admissible execution. In the calibrated instance with three slots the seven nonempty faces give $|H_7| = 7$.

Proposition 11.8 (Exhaustive partition of supports)**.** *For the seven nonempty faces of a triangle carrier, the* $2^7 = 128$ *supports are exhaustively partitioned by size into*

$$128 = 2 + 6 + 30 + 90,$$

with the 30 *refined to* $30 = 18 + 12$*, and classified into* 40 *orbits under* D_3 *and* 20 *archetypes under* $D_3 \times C_2^{\neg}$*.*

Status. An exhaustive finite enumeration of the declared fixture, reported as a calibration. The refinement of 30 into $18 + 12$ separates the chiral classes from the achiral ones. □

No-go 11.9 (Mask data does not recover direction)**.** The mask enumeration of Proposition 11.8 carries no construction direction: two constructions differing in orientation and agreeing on the mask are not separated by any mask-level datum.

Proof. The mask records which faces are nonempty, and the datum is invariant under the orientation-reversing involution of the carrier; a construction history is not. □

11.4 Fibre semantics

Definition 11.10 (Filling fibre)**.** For a formula φ, a model carrier $\mathcal{M}$, and an assignment ρ to the free slots, the *filling family* is

$$\mathrm{Fill}_{\mathcal{M},\rho}(\varphi) \subseteq \mathcal{M}$$

the set of terminal worlds in which φ is realised. The three semantic values are then read off structurally:

- *truth* is an inhabiting line: $\mathrm{Fill}_{\mathcal{M},\rho}(\varphi) \neq \varnothing$;
- *falsehood* is an exhaustively empty fibre, certified;
- *unresolved* is neither, within the budget.

Proposal 11.11 (Connectives act on fibres)**.** Connectives are operations on filling families rather than on truth values: conjunction is intersection of families, disjunction is union, and the quantifiers are the adjoints of Proposition 9.9. Consequently the semantics is determined by what it does to families, and the truth values are derived readings.

Remark 11.12. Proposition 11.11 is where Part II pays for Part I's discipline. If truth is primitive, falsehood and unresolved collapse. If the family is primitive, they do not, and the four statuses of Definition 11.5 are the honest taxonomy.

11.5 The geometric reading, and its conditionality

A finite observer can be presented geometrically, but only conditionally.

Proposal 11.13 (Finite observer as a marked surface)**.** A finite observer may be presented as a marked finite-type surface

$$\Sigma_P^\circ \simeq \Gamma_P \backslash \mathbb{H}^2_{\mathrm{AM}},$$

whose sealed threads are the vocabulary generators, with a cap–seal calculus and a J-operator acting on thread ends. In this presentation universality is an *axiom proposal*: the observer stores finitely many generators, relations, markings and residuals plus a rule producing any demanded finite portion, so infinity is potential rather than completed.

Warning 11.14 (Conditionality of the geometric reading). Proposition 11.13 is a representation claim, not an identification. The passage between the symbolic grammar and the hyperbolic carrier requires an explicit coding or quasi-isometry theorem, and none is supplied here. Statements about the hyperbolic model in this paper are imported from Paper 0 [3] and are not re-derived.

11.6 Summary

1. The typed hole is primitive, it denotes a filling family, and open and close are not inverses (Definition 11.2, Proposal 11.3).
2. Four statuses are needed, and exhaustion differs from unresolved (Definition 11.5, Warning 11.6).
3. Semantics is fibre-valued, with truth, certified falsehood, and unresolved kept apart (Definition 11.10, Proposal 11.11).
4. The geometric reading is available but conditional (Proposal 11.13, Warning 11.14).

12 Finite adequacy of threaded logic

Definition 11.10 gives a semantics. This section proves that between that semantics and a threaded derivability relation there is an exact adequacy, and it states the hypotheses on which the equivalence depends. The hypotheses are the content: each of them is a place where an unrestricted claim would fail.

12.1 The judgment

Definition 12.1 (Threaded judgment)**.** A *threaded judgment* is a sequent whose antecedents are *threaded*: each antecedent is attached to a thread, and threads are consumed by the rules rather than shared. Writing Γ for a threaded context and φ for a formula, we write $\Gamma \vdash \varphi$ for derivability in the threaded calculus.

The essential point is linearity of threads: no rule duplicates or discards an antecedent thread. Copy and discard, where they occur, are explicit connectives rather than structural rules. This is the syntactic shadow of Proposition 5.15: the fan-out must be written down.

Definition 12.2 (Realisation)**.** Given a carrier $\mathcal{M}$, an assignment ρ, and a threaded context Γ, a *realisation* of Γ is a choice, for each thread, of a terminal world compatible with the thread's formulas. Write $\mathcal{M}, \rho \models \Gamma$ when a realisation exists, and $\mathcal{M}, \rho \models \varphi$ when $\mathrm{Fill}_{\mathcal{M},\rho}(\varphi) \neq \varnothing$.

12.2 The four hypotheses

Definition 12.3 (Adequacy hypotheses)**.** The adequacy theorem below is stated under four hypotheses on the carrier and the connectives.

1. **Cartesian.** The carrier is a product of independent slot domains, so that a partial assignment extends to a terminal world whenever no constraint is violated.
2. **Compatibility.** The constraints attached to a threaded context are pair-wise compatible exactly when their conjunction is nonempty.
3. **Transformer.** Each connective has an associated transformer on filling families, and the transformer is sound and complete for the connective's semantics.
4. **Exhaustiveness.** For each atomic formula and each partial assignment, either a witness is produced or the emptiness of the family is certified.

Each hypothesis rules out a genuine failure mode. Without (1) a partial assignment need not extend; without (2) pairwise nonemptiness need not imply joint nonemptiness — which is exactly the obstruction of Proposition 9.11; without (3) a connective has no semantics on families; without (4) the fourth and third statuses of Definition 11.5 collapse.

12.3 The adequacy theorem

Theorem 12.4 (Threaded adequacy)**.** *Assume the four hypotheses of Definition 12.3. Then for every threaded context* Γ *and formula* φ*,*

$$\Gamma \vdash \varphi \quad \Longleftrightarrow \quad \text{for all } \mathcal{M}, \rho : \ \mathcal{M}, \rho \models \Gamma \Rightarrow \mathcal{M}, \rho \models \varphi.$$

Equivalently, in the fibre reading,

$$\mathcal{M}, \rho \models \varphi \iff \mathrm{Fill}_{\mathcal{M},\rho}(\varphi) \neq \varnothing.$$

Proof. By structural induction on derivations and on formulas.

Soundness direction. Suppose $\Gamma \vdash \varphi$ with a derivation π. We show each rule preserves realisation. For an assumption, the conclusion's thread is among the realised threads by Definition 12.2. For a conjunction rule, hypothesis (3) makes the transformer of $\wedge$ exactly intersection of families, so a realisation of both premises realises the conclusion. For a linear implication rule, hypothesis (1) allows the discharged thread to be replaced by the witness supplied for the premise, and the conclusion's family is nonempty because a witness exists for the implication itself. For quantifier rules, hypothesis (1) lets a witness for a free slot be chosen, and hypothesis (2) guarantees that the pairwise-compatible constraints are jointly satisfiable, which is what licenses the introduction rule.

Completeness direction. Suppose every realisation of Γ realises φ. Take the canonical assignment ρ^* built from the family data: for each slot choose, by hypothesis (4), either a witness or a certification of emptiness. If the family of φ were empty under every ρ realising Γ, hypothesis (4) would provide a certificate of emptiness; that certificate builds a derivation of the negated sequent. Since by assumption no realisation falsifies φ, the emptiness certificate cannot be produced for any realising ρ, so a derivation of $\Gamma \vdash \varphi$ exists by induction on the formula's construction. □

Calibration 12.5 (The theorem without the hypotheses)**.** For each of the four hypotheses there is a finite fixture on which the equivalence of Theorem 12.4 fails while the other three hold. The fixtures are the calibrations of this section; they are the reason the hypotheses are stated rather than assumed silently.

12.4 Substitution

Theorem 12.6 (Fibre-natural substitution)**.** *Under the hypotheses of Definition 12.3, substitution of a derivable formula for a free slot commutes with the filling-family construction:*

$$\mathrm{Fill}_{\mathcal{M},\rho}(\varphi[\psi/x]) = \mathrm{Fill}_{\mathcal{M},\rho[x\mapsto \mathrm{Fill}_{\mathcal{M},\rho}(\psi)]}(\varphi).$$

Proof. Induction on φ. At an atom the two sides are the same family by definition of the extended assignment. At a connective, hypothesis (3) makes the transformer act on families, and the induction hypothesis identifies the substituted families on each side; the transformer's functoriality then gives the identity at the composite. □

Remark 12.7. Theorem 12.6 is the semantic counterpart of the syntactic substitution theorem of §14. The two are not the same statement: the present one is about families, the later one is about ledgers.

12.5 Promotion boundary

Calibration 12.8 (What this section does not license)**.** Theorem 12.4 is a finite semantic theorem about a declared carrier and a declared connective set. It licenses no claim of the following forms: that the calculus is complete for an unrestricted semantics; that the four hypotheses are necessary; that quantifiers behave classically; or that the carrier's finiteness is an artefact of presentation. In particular the theorem is stated for Boolean semantics on 2^D, hence in particular on H_D, and says nothing about carriers with dependent slot domains.

12.6 Summary

1. Threaded judgments consume threads linearly, so fan-out is explicit (Definition 12.1).
2. Adequacy holds under four named hypotheses, each of which blocks a genuine failure (Definition 12.3, Theorem 12.4).
3. Substitution is fibre-natural (Theorem 12.6).
4. The theorem's scope is finite and declared (Calibration 12.8).

13 Ordered linear deduction: soundness and its limit

§12 gave a semantics and a calculus adequate for it under hypotheses. This section fixes the specific fragment studied in the rest of Part II and asks the two questions that matter about it: is it sound, and is it complete? The answers are yes and no, and the failure of completeness is exact rather than mysterious: it is measured by a grade invariant.

13.1 The fragment

Definition 13.1 (The ordered linear fragment TND_0)**.** TND_0 is the implicational fragment of threaded natural deduction in which

1. antecedents form an *ordered* context: order is part of the judgment and there is no exchange;
2. there is no weakening and no contraction; copy and discard, if present, are explicit terms;
3. a judgment carries a *ledger* recording which antecedents each conclusion consumed.

Definition 13.2 (Ledger-indexed judgment)**.** Write

$$L \mid \Gamma \vdash \varphi$$

for the judgment that φ is derivable from the ordered context Γ while consuming exactly the ledger L. The ledger is not notation for the context: two judgments may share Γ and differ in L.

The distinction between Γ and L is the syntactic analogue of the distinction between source and occurrence that Part I insisted on (Proposition 5.15). A context lists what is available; a ledger records what was used, and by which use.

13.2 Soundness

Theorem 13.3 (Soundness of TND_0)**.** *If $L \mid \Gamma \vdash \varphi$ then for every carrier $\mathcal{M}$ and assignment ρ,*

$$\mathcal{M}, \rho \models \Gamma \;\Rightarrow\; \mathcal{M}, \rho \models \varphi.$$

In particular TND_0 *is sound for Boolean semantics on 2^D and hence on the terminal-world carrier H_D of Definition 11.7.*

Proof. Induction on the derivation, using only that the fragment's rules preserve realisation of the ordered context: the implication-introduction rule discharges the head of the ordered context, implication-elimination consumes it, and no rule moves antecedents. The argument is the ordered specialisation of Theorem 12.4 and does not need hypotheses (1)–(3) beyond the product structure of 2^D. □

13.3 The grade invariant, and incompleteness

Definition 13.4 (Grade)**.** For a formula φ and a terminal world assignment, let $\mathrm{gr}(\varphi)$ be the alternation count of the leading connective along the leftmost spine, read as a natural number. For a derivation, gr is the maximum over its judgments.

Theorem 13.5 (Incompleteness)**.** TND_0 *is* not *complete for Boolean semantics on 2^D: there exist Γ and φ with $\mathcal{M}, \rho \models \Gamma \Rightarrow \mathcal{M}, \rho \models \varphi$ for all ρ, yet $L \mid \Gamma \vdash \varphi$ is underivable for every ledger L.*

Proof. Semantic entailment on 2^D is closed under the classical structural rules. Take a semantically valid implication whose witnessing term requires a permutation of antecedents (exchange) or the reuse of one antecedent (contraction). By Definition 13.1 neither is available, and by Definition 13.2 a ledger cannot be shared between two uses. Hence no derivation exists, although the entailment holds. □

Proposition 13.6 (The grade measures the gap)**.** *The mismatch between Boolean entailment and threaded derivability is measured by the grade of Definition 13.4: for a fixed carrier, the minimal grade at which a valid entailment becomes derivable is a computable function of the formula, and it is unbounded over formulas.*

Status. The computation is finite on each formula by the decision procedure of §15; unboundedness is exhibited by the family of formulas requiring n nested exchanges. □

Remark 13.7 (Why incompleteness is a result, not a defect)*.* Theorem 13.5 tells us that the ordered linear fragment *selects* a logic strictly finer than Boolean entailment. On the reading of Proposal 9.13 that is information about which logic the grammar encodes, not a bug to be repaired by adding structural rules.

13.4 Triadic conflict apertures

The fragment has one genuinely triadic rule, and it is worth isolating.

Proposal 13.8 (Conflict aperture)**.** Suppose two lines of derivation carry distinct tri-domain identities and compete for one linear source. Then the admissible aperture is uniquely determined by the complement of the two occupied domains:

$$\mu(d, e) = D \setminus \{d, e\},$$

and a triangular cell witnessing the conjunction $\wedge_{\triangle}$ exists exactly when a certified linear common lift of the two lines is available.

Proposition 13.9 (Only a certified common lift closes the cell)**.** *If the two lines have no certified linear common lift, then the triangular cell of Proposal 13.8 admits no filler in* TND_0*, and the aperture remains in the* unresolved *status of Definition 11.5.*

Proof. Opening the aperture consumes the source once, by linearity; closing it requires two inhabitants of the complementary domain that agree on the source. Without a certified common lift the two inhabitants are not related by any derivation of the fragment, so no filler exists. Since the search space is finite and the fragment is decidable (§15), failure is reported as unresolved rather than as falsity, in accordance with Warning 11.6. □

Remark 13.10. The word "triadic" here is doing precise work. The three domains $d, e, \mu(d, e)$ are not three independent coordinates: once two are occupied the third is determined. This is the syntactic counterpart of Proposition 7.5, and, like it, it does not by itself produce a non-decomposable triple defect (Calibration 9.12).

13.5 Summary

1. TND_0 is ordered, linear, and ledger-indexed (Definitions 13.1, 13.2).
2. It is sound for Boolean semantics on 2^D and hence on H_D (Theorem 13.3).
3. It is provably incomplete, and the gap is measured by a grade (Theorem 13.5, Proposition 13.6).
4. Its one triadic rule yields a uniquely determined aperture whose cell closes exactly under a certified common lift (Proposal 13.8, Proposition 13.9).

14 Substitution, ledger transport, and strong normalization

The proof theory of TND_0 begins with substitution, because an ordered linear fragment cannot borrow the usual substitution lemma: the order is part of the data, and the ledger records uses that an ordinary substitution would silently move. This section proves an ordered one-hole substitution theorem with an exact ledger splice, derives conditional cut admissibility from it, shows that raw ledger equality is the wrong target, and then transports the whole apparatus through proof contexts to obtain strong normalization.

14.1 Ordered one-hole substitution

Definition 14.1 (One-hole context)**.** A *one-hole proof context* is a proof tree with a distinguished hole at a leaf. Two proof trees are *concatenable* along the hole when the type of the conclusion at the hole matches the type of the root of the inserted tree; concatenation is order-respecting and is written $Q[\cdot]$.

Definition 14.2 (Ordered substitution)**.** Let σ be a substitution assigning to each free slot of the antecedent block a position in a target context. The *ordered substitution* $P\sigma$ is obtained by replacing each free occurrence in order of the context, without permutation.

Theorem 14.3 (Ordered substitution with ledger splice)**.** *Let Λ_σ be the ledger change induced by the substitution σ. For a one-hole context $L\,Q\,R$ in which Q sits between a left ledger block L and a right ledger block R,*

$$\mathrm{ledger}(L\,\Lambda_\sigma\,R) = L \circ \Lambda_\sigma \circ R,$$

an exact splice: the left block is untouched, the middle block is transported by σ, and the right block is untouched, with no reordering.

Proof. Induction on the context $L\,Q\,R$. At the hole the ledger is exactly Λ_σ by Definition 14.2. For each rule above or below the hole, the rule's own ledger contribution is a block that does not commute with the hole's block, which is the content of linearity: the induction hypothesis is applied to the strict subcontexts, and the blocks compose in the order fixed by the derivation. □

Corollary 14.4 (Fresh-cut admissibility, conditionally)**.** *If the transported ledger Λ_σ is* fresh*, meaning no antecedent is consumed twice, then the cut on the substituted formula is admissible: the composite $L\,\Lambda_\sigma\,R$ is derivable whenever the substitutes are.*

Proof. Apply Theorem 14.3 and check that each antecedent in the spliced ledger occurs exactly once. Freshness is exactly the condition that no contraction is implicitly required, and without it the splice would duplicate an antecedent in violation of Definition 13.1(2). □

Remark 14.5 (The condition is not removable). Corollary 14.4 is conditional and stays conditional. Dropping freshness would allow a ledger with a repeated antecedent, which no rule of the fragment can introduce; the resulting "admissible" cut would have no derivation.

14.2 Raw ledger equality fails

No-go 14.6 (Ledger equality is not the right invariant)**.** Raw equality of ledgers is not preserved by beta-like reduction: there are a proof tree and its reduct whose ledgers are unequal as words while both record the same current authority. Consequently the target of the proof transformation is not ledger equality but current ledger equality plus a certificate for the retired part.

Proof. A principal-implication detour consumes the antecedent in one order and its reduct consumes it in another; both consume it exactly once. The ledgers therefore differ as words while agreeing in the multiset of current antecedents. □

Proposal 14.7 (Ledger transport, not ledger identity)**.** The correct statement is a *transport*: a reduction relates the pre-state ledger to the post-state ledger by a declared transformation, and retired authority leaves the current ledger while surviving in the certificate. This is controlled forgetting in the sense of Proposition 9.13, not equality.

14.3 Transport through contexts

Lemma 14.8 (Freshness reflects through discharge)**.** *If a ledger is fresh above a discharge, then its image under the discharge is fresh, and conversely a fresh image implies freshness above whenever the discharge consumes exactly one antecedent.*

Proof. The discharge removes exactly one antecedent occurrence and appends the discharged assumption to the premise's right block. Freshness is a condition on occurrences, so it is preserved in both directions under this exact-one bookkeeping. □

Theorem 14.9 (Root transport lifts through every context)**.** *Let*

$$L\,[x^b]\,Q\,S \;\rightsquigarrow\; L\,S\,Q$$

be the root beta-like transport of a principal detour. Then for every one-hole proof context $C[\cdot]$,

$$C(L\,[x^b]\,Q\,S) \;\rightsquigarrow\; C(L\,S\,Q),$$

and the transported ledger differs from the original by the declared transformation only.

Proof. By induction on the context, using Lemma 14.8 at each discharge and Theorem 14.3 at each substitution. Linearity ensures that no context can duplicate the detour, so the transport is well defined at every depth. □

Corollary 14.10 (Contextual closure)**.** *The beta-like reduction relation of* TND_0 *is closed under one-hole contexts.*

Proof. Immediate from Theorem 14.9. □

14.4 Strong normalization

Definition 14.11 (The two measures)**.** For a proof tree P let $\mu_1(P)$ be the number of principal-implication detours and let $\mu_2(P)$ be the total size of the ledgers above the detours. Order pairs $(\mu_1(P), \mu_2(P))$ lexicographically.

Theorem 14.12 (Strong normalization)**.** *Every reduction sequence in* TND_0 *terminates.*

Proof. By Theorem 14.9 a reduction step may be applied at any depth, so it suffices to show that each step strictly decreases the lexicographic pair of Definition 14.11. A root transport removes one detour, hence decreases μ_1 by exactly one; the ledgers of the transported subterms are unchanged as words by Theorem 14.3, so μ_2 cannot increase. A step inside a context leaves μ_1 fixed only if no detour is removed there, in which case it acts on a strictly smaller tree and the induction on tree size applies; otherwise μ_1 decreases. Since $\mathbb{N} \times \mathbb{N}$ is well ordered under the lexicographic order, no infinite sequence exists. □

Remark 14.13 (Where linearity was used). Theorem 14.12 depends on linearity twice: in Theorem 14.3, which forbids reordering the ledger, and in Lemma 14.8, which keeps freshness intact through discharge. The same argument fails for a fragment with contraction, because μ_1 need not decrease when one detour is copied into several.

14.5 Summary

1. Ordered substitution produces an exact ledger splice (Theorem 14.3).
2. Cut admissibility is conditional on freshness (Corollary 14.4), and the condition is not removable.
3. Ledger equality is the wrong invariant; transport plus certificate is right (No-go 14.6, Proposal 14.7).
4. Root transport lifts through all contexts, and reduction is strongly normalizing (Theorem 14.9, Corollary 14.10, Theorem 14.12).

15 Confluence and decidable derivability

Normalization in §14 guarantees that a normal form exists. It does not guarantee that the normal form is unique, and uniqueness is what a decision procedure needs. This section supplies the critical-pair analysis that yields confluence, then the normal/neutral characterization that yields the subformula property, and finally the focused decision procedure that follows. It closes by describing precisely what finite decidability does *not* give.

15.1 Critical pairs

Because TND_0 is linear and ordered, the only way two redexes can interfere is that their ledgers overlap. This makes the critical-pair analysis finite and explicit.

Definition 15.1 (Overlapping redexes)**.** Two redexes in a TND_0 proof tree *overlap* when their ledgers share an antecedent occurrence. Since ledgers are consumed linearly (Definition 13.1), a shared occurrence means one redex lies on the path from the other to its discharged assumption.

Proposition 15.2 (Four-way path classification)**.** *For an overlapping pair of redexes, exactly one of four configurations occurs:*

1. *one redex strictly above the other, on the discharged spine;*
2. *one redex strictly above the other, off the discharged spine;*
3. *the two redexes are adjacent along one ledger block;*
4. *the two redexes are disjoint but order-adjacent in the ledger.*

Proof. By Definition 15.1 the two redexes lie on one spine, and the relative position of the shared occurrence in their ledger blocks gives the four cases. Linearity excludes any configuration in which a ledger block is shared without one redex containing the other. □

Lemma 15.3 (Substitution and graft equations)**.** *In configurations (1) and (3) the two reduction orders are related by the ledger splice of Theorem 14.3; in configurations (2) and (4) they commute exactly.*

Proof. Configurations (2) and (4) act on disjoint ledger blocks, so the root transports of Theorem 14.9 commute by inspection. Configurations (1) and (3) nest one substitution inside the other, and the splice law identifies the two composites; the only thing to check is that the ledger blocks compose in the same order in both, which is the content of freshness. □

15.2 The one-step strong diamond

Theorem 15.4 (One-step strong diamond)**.** *If $P \rightsquigarrow P_1$ and $P \rightsquigarrow P_2$ with $P_1 \neq P_2$, then there is P_3 with $P_1 \rightsquigarrow P_3$ and $P_2 \rightsquigarrow P_3$, and the two-step reductions can be chosen to have equal length.*

Proof. By Lemma 15.3, each configuration of Proposition 15.2 either commutes or is joined after one further step. The equal-length clause follows because both reductions remove the same number of detours in configurations (2) and (4), and differ by a single commuting step in (1) and (3). □

Corollary 15.5 (Local and global confluence)**.** *The reduction relation of* TND_0 *is locally confluent and, being strongly normalizing (Theorem 14.12), globally confluent by Newman's lemma. Consequently every proof has a unique normal form.*

Remark 15.6. Corollary 15.5 is stated for a *fixed* proof tree. It does not say that two distinct proofs of the same sequent share a normal form; the fragment is incomplete (Theorem 13.5), so distinct derivations of one sequent need not be related at all.

15.3 Confluence is not coherence

No-go 15.7 (Confluence does not give a 2-cell structure)**.** The fact that reductions commute is not the fact that they are coherent. There is a four-level ladder

$$\text{proof endpoint} \subsetneq \text{composite boundary transport} \subsetneq \text{history} \subsetneq \text{history 2-cell},$$

and confluence at the first level determines nothing at the fourth. Reading Corollary 15.5 as a coherence theorem is therefore invalid.

Proof. Levels 1 and 2 are quotients of level 3, and level 4 records relations among histories that are collapsed by the quotient. Confluence is a statement about the image at level 1; a nontrivial 2-cell lives in the kernel of the forgetful map from level 3 to level 1, and that kernel is not determined by level 1. □

Remark 15.8. No-go 15.7 is the proof-theoretic counterpart of Proposition 2.14. In both cases the error is the same: taking a quotient for the object.

15.4 Normal and neutral forms

Definition 15.9 (Normal and neutral)**.** A derivation is *normal* if it contains no principal-implication detour. It is *neutral* if it ends in an elimination whose major premise is a hypothesis. The standard characterization is: normal derivations end either in an introduction or are neutral, and neutral derivations are closed under the eliminations of the fragment.

Theorem 15.10 (Subformula property)**.** *Every formula occurring in a normal* TND_0 *derivation is a subformula of the end sequent or of a discharged hypothesis.*

Proof. Induction on the normal derivation using Definition 15.9. An introduction rule's premises contain only subformulas of its conclusion; an elimination whose major premise is neutral has a minor premise that is a subformula of the major premise's type, and the absence of detours means no formula is introduced only to be eliminated. □

Proposition 15.11 (Leftmost-head canonicity)**.** *In a normal derivation of an implicational sequent, the construction of the proof is determined by the ordered context together with the leftmost hypothesis of the end sequent's antecedent block. Hence the leftmost-head strategy is complete for the fragment.*

Proof. By Theorem 15.10 every formula of a normal derivation is a subformula of the goal, and by linearity no antecedent is reused, so at each stage the possible rules are determined by the leftmost unconsumed antecedent and the goal. The choices are therefore forced up to the order in which independent threads are processed. □

15.5 Decidable derivability

Theorem 15.12 (Focused decision procedure)**.** *Derivability in* TND_0 *is decidable. There is a terminating, sound, and complete procedure: search normal derivations in leftmost-head order, using Theorem 15.10 to bound the formula set, Definition 15.9 to alternate focused and unfocused phases, and Proposition 15.11 to fix the branching.*

Proof. Termination: by Theorem 15.10 only finitely many formulas may occur, and linearity bounds the number of introductions by the number of antecedents. Soundness: each step of the procedure builds a derivation, so a found proof is a proof. Completeness: by Corollary 15.5 a normal form exists if any derivation does, and by Proposition 15.11 the procedure explores a branch containing it. □

Proposition 15.13 (Positioning of the skeleton)**.** *The* TND_0 *skeleton is the associative, empty-antecedent, right-residual fragment of ordered linear logic. Under this reading, the erasure of audited nominal data to a skeleton is a functor, and lifting back is by fresh, noncanonical decoration; long forms are not identified with their eta-expansions, and multiple focused proofs of one sequent are admitted.*

Status. The fragment identification is a syntactic matching of rule shapes. Proposition 15.11 and Theorem 15.12 are the substantive content; the erasure/lift discipline is a definitional choice, recorded so that the decision procedure's non-canonicity of decoration is visible rather than implicit. □

15.6 What finite decidability does not give

Warning 15.14 (No compactification from finite search)*.* Theorem 15.12 is a statement about a finite fragment. It does not produce, and must not be read as producing, a halting measure, a limit object, or any Ω-type boundary. Neither finite confluence (Corollary 15.5) nor finite focused search (Theorem 15.12) yields such an object; they are finite statements and the boundary is not their limit.

Remark 15.15*.* Warning 15.14 is the sharpest instance of the discipline that organises this paper. A decision procedure terminates; that is not the same as exhibiting a limit of a non-terminating process, and conflating the two would convert a finite theorem into an infinite claim.

15.7 Summary

1. The critical-pair analysis is finite, with four configurations (Proposition 15.2, Lemma 15.3).
2. Reduction satisfies a one-step strong diamond and is confluent, so normal forms are unique per proof (Theorem 15.4, Corollary 15.5).
3. Confluence is not coherence (No-go 15.7).
4. The subformula property plus leftmost-head canonicity give a terminating, sound, and complete decision procedure (Theorems 15.10, 15.12, Proposition 15.11).
5. Decidability yields no Ω-type boundary (Warning 15.14).

16 The six-port whole-cut kernel

Part II has so far supplied a semantics, a calculus, and a metatheory. This section asks how few ports a carrier needs in order to express that calculus with three readings and binary cuts, and answers with a counting theorem. The answer is six. The section is careful about what the number means: it is a conditional minimum under a stated contract, not an absolute truth, and the universality claim that usually accompanies such a number is replaced by relative initiality.

16.1 The counting question

Definition 16.1 (Whole-cut carrier)**.** A *whole-cut carrier* with m domains and n ports consists of

1. a finite set D of domains;
2. for each domain, a binary cut $\langle L \mid R \rangle$ that is an atomic generator rather than a derived construct;
3. a set P of ports, each attached to one domain and one side of its cut;
4. an occurrence-and-multiplicity ledger M recording, for every term, the production and consumption of each occurrence;
5. a routing datum γ, a history ledger H, and a residual record R.

Definition 16.2 (The contract)**.** The counting theorem is stated under six clauses.

1. **Atomic cuts.** Each cut is a generator; it is not decomposed into smaller structure.
2. **Binary sides.** Each cut has exactly two complementary sides.
3. **Two-sided access.** Every domain must be readable on both sides of its own cut, since a view that could read one side only could not express a crossing.
4. **Occurrence fidelity.** No view, braid, collision, projection, or quotient may create, drop, rename, or merge an occurrence.
5. **Production witnesses.** Copies require production witnesses; collisions conserve multiplicity.
6. **View discipline.** Every view extends the ledger and commutes with the boundary operator: $V_\nu(\partial c) = \partial V_\nu(c)$.

16.2 The minimality theorem

Theorem 16.3 (Counting bound)**.** *Under the contract of Definition 16.2, a triadic whole-cut carrier with binary cuts satisfies*

$$n \;\geq\; 2m,$$

and hence, for $m = 3$, $n_{\min} = 6$.

Proof. By clause (3) every domain must be accessible on both sides of its own cut, so each domain contributes at least two distinct ports; by clause (1) the cut is atomic, so those two ports cannot be served by sub-structure; by clause (2) the two sides are complementary, so a single port cannot serve both. Hence $n \geq 2m$, with equality attained by the carrier $\mathbb{W}_6$ of §16.3. □

Remark 16.4 (What the proof does and does not use). The bound uses clauses (1)–(3) only; clauses (4)–(6) are needed for the carrier to be usable, not for the count. This split matters, because it isolates exactly which part of the contract is under pressure if the number six is ever challenged.

Warning 16.5 (Six is conditional). Theorem 16.3 says that six is minimal *for this contract*. Drop clause (3) and a carrier with three ports becomes available; strengthen clause (1) to require decomposed cuts and the minimum changes again. The number six is therefore a statement about the contract, and reporting it as an absolute minimality of any three-domain syntax would be an error.

16.3 The carrier

Definition 16.6 (The six-port carrier)**.** The carrier is

$$\mathbb{W}_6 = (\mathsf{D},\ C,\ P,\ O,\ M,\ \gamma,\ H,\ R)$$

consisting of domains, cuts, ports, occurrences, the multiplicity ledger, routing, history, and the residual record. Its distinguishing structural features are:

1. a *split pair*, giving the two sides of each atomic cut;
2. a *projector*, identifying which ports a given view reads;
3. a *factored traversal*, so that a traversal decomposes along the cut rather than cutting across it;
4. *collision strata*, recording where two occurrences coincide.

Proposal 16.7 (Line and circle are views)**.** In $\mathbb{W}_6$ the *line* and the *circle* are not neighbouring value constructors but checked projections of one ledger, each with a declared kernel. One atomic cut co-generates a continuation germ and a boundary germ; the line reads the first and the circle the second.

Status. A definitional restructuring, justified by clause (6): both readings are views V_ν of the same boundary operator, so neither is primitive. The claim asserted is the commutation $V_\nu(\partial c) = \partial V_\nu(c)$, which holds by construction of the two views and is checked on the finite fixture. □

16.4 The grammar, the ledgers, and the degree

Definition 16.8 (Whole-cut record)**.** A *whole-cut record* consists of port, cut, and cell records together with six ledgers: *production*, *through*, *cycle*, *hole*, *alternative*, and *residual*. The record carries a *degree* $\deg(W)$, and for the minimal carrier $\deg(\mathbb{W}_6) = 6$.

Proposition 16.9 (Collision conservation)**.** *In any well-typed whole-cut record the multiplicity ledger is conserved under every rule of the grammar: no production is created or destroyed, and every collision accounts for exactly one coincidence of two occurrences.*

Proof. Each grammar rule consumes and produces ledger entries in matching multiplicity; the production ledger records copies explicitly, so a rule that duplicates a term must record a production, and clause (5) forbids doing so silently. Induction over derivations in the grammar. □

Remark 16.10 (Distributivity as a recorded witness). The grammar's distributivity step is accompanied by an explicit witness record with the ordered hole bindings. The content is not the equation — which holds in the intended model — but the record of which occurrences of the duplicated term were produced, in which order. Without that record the equation would silently license a contraction, which No-go 14.6 and Proposition 16.9 forbid.

16.5 The cell–carrier–view spine

Proposal 16.11 (Spine)**.** There is a typed spine

$$\mathsf{TypedCellSignature} \xrightarrow{\mathsf{assemble}_\rho} \mathsf{StratifiedCarrier} \xrightarrow{V_\nu} \mathsf{PresentedView}_\nu,$$

in which the first arrow assembles cells into a stratified carrier along a declared stratification ρ, and the second is the view of clause (6). The boundary operators of the two basic cells are

$$\partial\mathsf{InterchangeCell} = Q_4, \qquad \partial\mathsf{BraidCell} = M_6,$$

with Coxeter shadows $C_2 \times C_2$ and S_3 respectively.

Status. The boundary identifications are computations in the cell complex: an interchange cell is a square whose boundary reads as a product of two transpositions, a braid cell a hexagon whose boundary reads as the three-strand braid relation. The Coxeter shadows are the standard reflections of the square and hexagon. □

Proposition 16.12 (Envelope count)**.** *The truncated-octahedral envelope of the stratum satisfies*

$$TO_{24} = 6\,Q_4 + 8\,M_6,$$

that is, six squares and eight hexagons on twenty-four vertices, and the relation boundary count is $2m$ *in the sense of Theorem 16.3.*

Proof. $6 \cdot 4 + 8 \cdot 6 = 72$ face–vertex incidences on 24 vertices with degree 3: $24 \cdot 3 = 72$. The consistency of the two counts witnesses that the envelope closes. □

16.6 The M_6 boundary bridge

Theorem 16.13 (Alternating-pairing bridge)**.** *Let κ and τ be two fixed-point-free involutions on the six-port ledger whose union is connected. Then the union of their graphs is a two-regular graph on six vertices, hence a six-cycle C_6, whose edges alternate between the two kinds. Consequently the checked incidence data present the open boundary*

$$\mathsf{bridge}(\mathbb{W}_6) = M_6^{\partial}\big[\kappa\tau\kappa \Rightarrow \tau\kappa\tau\big],$$

with no filler supplied.

Proof. A fixed-point-free involution on six elements is a perfect matching, so it contributes three edges. Two matchings whose union is connected give a two-regular graph, which is a disjoint union of cycles; connectivity forces a single six-cycle. Alternation of edge kinds follows from the two matchings being distinct, and the displayed presentation of M_6^{∂} records the two alternating readings of the cycle. The absence of a filler is Calibration 16.14. □

Calibration 16.14 (The filler is open)**.** The bridge of Theorem 16.13 presents the boundary M_6^{∂} as an open object: the two alternating readings are exhibited, and no 2-cell filling them is constructed. Every construction in this material that touches M_6 lists the filler as an open residual.

16.7 Relative initiality instead of universality

Proposal 16.15 (Relative initiality)**.** $\mathbb{W}_6$ is initial in the category of presented models satisfying the contract of Definition 16.2: for every such model N there is a morphism $\mathbb{W}_6 \to N$, unique up to the declared decoration.

Status. The statement is a Proposal. What is available is the counting argument of Theorem 16.3 together with the view discipline of clause (6), which supply the candidate morphisms; no uniqueness proof is given here. □

Warning 16.16 (Not a universality theorem). Proposition 16.15 replaces, and does not establish, a claim of universality. In particular it says nothing about computation-universality, and it does not assert that every three-domain syntax factors through $\mathbb{W}_6$. The distinction is the departure point of §17.

16.8 Summary

1. Six is the minimum under a stated six-clause contract (Definitions 16.1, 16.2, Theorem 16.3, Warning 16.5).
2. In the minimal carrier, line and circle are views of one ledger (Definition 16.6, Proposal 16.7).
3. The ledgers conserve multiplicity, so distributivity must record its productions (Proposition 16.9).
4. The spine and envelope counts are explicit (Proposal 16.11, Proposition 16.12).
5. The M_6 boundary is presented with an open filler, and the kernel's status is relative initiality, not universality (Theorem 16.13, Calibration 16.14, Proposal 16.15, Warning 16.16).

17 Claim boundaries and the remaining programme

This paper has reported more boundary results than theorems, and the imbalance is deliberate. This section collects the boundaries, states which of them are open problems rather than exclusions, and identifies the single gap on which the triadic programme depends.

17.1 The decisive gap: order from energy

Every layer of Part I that computes a quantity — period grades, spectral packets, energy vectors, trace counts — is a map into something ordered or something measurable. None of them produces the structure that a cut actually needs.

Open problem 17.1 (Order from a finite frontier). Construct a chart-covariant certified comparison on finite period-energy frontiers whose objectification yields the correct order relation on the resulting numbers.

Remark 17.2 (Why this is the decisive gap). Energy and period data determine *what is visible*. Nothing in this material determines *what is left and what is right*. The identification of energy grades with the two sides of a cut was explicitly rejected as unlicensed (No-go 9.18 shows that energy alone is vacuous as a selector, and Proposition 9.17 that it detects only invariance). Until Problem 17.1 is solved, the triadic programme has carriers and residuals but no order, and by Definition 9.1 a logic without order has evidence but no entailment.

Remark 17.3 (The decisive gap is not a shortage of technique). Three no-go results of this paper already remove the obvious candidate classes of ingredient for the order that Problem 17.1 asks for. No-go 9.18 shows that energy alone is vacuous as a selector; Proposition 9.17 shows that what an energy detects is invariance; and No-go 11.9 shows that mask data, being invariant under the orientation-reversing involution of the carrier, carries no direction at all. Each of these is a scalar or finitely-valued functional of the carrier's state, and each is invariant under the symmetry that would have to be broken. The gap is therefore not the absence of a clever enough functional; what is missing is a datum of a different kind. We do not assert what that kind is, and this remark proves nothing beyond the three cited no-gos; we record it so that the programme's first task is not pursued by searching for another scalar.

17.2 Open problems

Open problem 17.4 (Canonical identification of the three readings). Decide whether the three readings $\rho_t, \rho_X, \rho_{\mathcal{K}}$ admit a canonical identification on the Legendre carrier, or whether only the residual-laden comparison of Theorem 7.15 exists. The mod-two description (Proposition 7.5) shows that at least the sign must be tracked; whether anything further is canonical is open.

Open problem 17.5 (Non-decomposable triadic defect). Prove or refute Conjecture 7.20: does there exist a carrier on which the cyclic defect of the three crossings is not reconstructible from any pair? The finite Legendre prism fails (Theorem 7.17), and the finite satisfaction form fails (Calibration 9.12); both failures are decompositions of a specific kind, and neither excludes a carrier of a different kind.

Open problem 17.6 (The M_6 filler). Construct the 2-cell that fills M_6^{∂} as presented by Theorem 16.13, or prove that the alternating presentation admits no filler. The boundary is exhibited; the interior is not.

Open problem 17.7 (A typed feedback operator). Supply a syntactic operator whose semantics is genuinely periodic, thereby repairing No-go 8.21. Boundary-return words exist; periodic dynamics does not. The obstruction is that completed-past rank is monotone, so any repair must change the carrier rather than add a formula.

Open problem 17.8 (Truncation is residual-valued). Finite energy sublevels are not closed under addition, multiplication, or grafting, so truncation satisfies $xy = P_\Lambda(xy) + R_\Lambda(x, y)$ with nonvanishing residual. Give a calculus in which this residual composes.

Open problem 17.9 (Vocabulary gradients). For a residual functional E on monotone Boolean vocabularies, the diminishing-returns inequality holds, and the heuristic "largest local gain is best" is refuted by explicit counterexample. Determine the correct admissible search discipline, and connect it to the coaction of Proposition 6.8.

Open problem 17.10 (Learning and proof as reverse duals). The duality proposed in this material is contravariant: proof normalises toward a sink and learning is the reverse generative orientation, with the forward equation false. Supply the formal dual pair and prove that the two orientations are exchanged, or show that only one direction exists.

17.3 What is excluded, not merely open

The following are excluded by scope, and no amount of further work inside this paper would include them.

1. **Universality claims.** The kernel's status is relative initiality (Proposition 16.15, Warning 16.16). Computation-universality of the triadic form is not claimed at any point.

2. **Physical interpretation.** The six-state carrier is a finite combinatorial object (Theorem 4.9); spacetime, metric signature, and physical realization are outside the paper. Warning 4.7 removes the most tempting identification explicitly.

3. **Complex scalars as a foundational necessity.** What is proved is a selection within a tested family under a bridge hypothesis (Proposition 3.6, Calibration 3.13).

4. **Computational lower bounds.** No growth rate, no entropy inequality, and no fibre size in this paper yields a runtime or space bound. Paper IV owns that territory under stricter hypotheses [5].

5. **Ω-type boundaries from finite results.** Finite confluence and finite focused search produce no limit object (Warning 15.14).

6. **Knot-theoretic invariance.** The figure-eight calibration (Theorem 8.19) identifies one feature of one word; it does not construct a knot invariant, and the polynomial does not reconstruct the knot.

17.4 Engineering excluded from the paper

The governance scope of this paper (`governance/00c-paper-V-triadic-aperture-scope.md`) excludes the engineering scaffolding of the source corpus: checked-program harnesses, test suites, authority splits, work-package phases, document formats, ledgers of artifacts, and CI gates. Fifteen of the 133 surveyed source files contribute no main-text mathematics and are cited at most as provenance. Where a mathematical claim in this paper depends on an executed check, the dependence is recorded in the claim ledger of Appendix D rather than concealed.

Appendix D also lists the source clusters that are mathematical but belong to neither part of this paper — the causal-cut and Alexandrov-duality notes, the backward transport and cut-composition notes, the PSP^* programme, the Ω-boundary cluster, and the later formation-gate notes. Those omissions are deliberate and are recorded so that they are visible rather than accidental; they are the natural content of a subsequent paper.

17.5 The programme

Four tasks follow, in order of logical dependence.

1. **Order.** Solve Problem 17.1. Nothing downstream is stable until a comparison is available, because a logic without order is only evidence.

2. **Filler.** Solve Problem 17.6, which completes the kernel's presentation and would decide whether the triadic structure is carried by the cell or only by its boundary.

3. **Triadicity.** Settle Problem 17.5, which decides whether the paper's central negative calibration is a limitation of the finite models or a feature of the phenomenon.

4. **Motion.** Solve Problem 17.7, which is what would turn the syntax of Part II into a semantics with dynamics.

Remark 17.11 (On the shape of the programme). Three of the four tasks are constructive and one is a decision. That ratio is the honest summary of the state of this material: the carriers and the residuals exist, and what is missing is the additional structure that would make them a theory of order and motion.

18 Conclusion

This paper set out to ask what structure is needed for one carrier to be read three ways, what such readings necessarily lose, and where the loss can be written down and checked. The answers came in two parts.

Part I produced a theory of characteristics in which the primitive is an observer contract and the product is a quotient together with a residual. Within that framework three results are stable. The quadratic trichotomy shows that no coarse requirement selects the complex field, and that what selects it is the retention of a central sign — chirality, isolated as the datum distinguishing the two lifts of one projective exchange. The six-state carrier is not postulated but generated: cut-membership cochains and oriented interchange produce the Clifford relation on a declared local cut interval, and interchange alone provably cannot. And crossing is a cospan with a resolution torsor, from which monodromy is recovered as a difference of resolutions and the Legendre monodromy follows from a quadratic base change.

Part I also produced a systematic catalogue of shadows that do not determine construction — spectra, gradients, snapshots, objectified values, masses, counts, windings, energies — and showed that the repair is never a finer scalar but always a residual. Where a residual is supplied the theory becomes faithful: marked jets and declared port charts are the two worked instances. One obstruction in the catalogue comes with an exact minimum (Theorem 10.18), and that theorem is the standard the others have not yet met.

Part II moved from what a reading cannot see to the places where it records the absence. The typed aperture is primitive, its open and close are not inverse, and four statuses are required because exhaustion and unresolvedness are different things. On that primitive a finite semantics, a threaded adequacy theorem under four named hypotheses, an ordered linear calculus that is sound but provably incomplete, and a metatheory were built: an ordered substitution theorem with an exact ledger splice, conditional fresh-cut admissibility, strong normalization via two decreasing measures, confluence with unique normal forms, the subformula property, and a terminating and complete focused decision procedure. The last part of the paper answered the counting question for the kernel: six ports, under a stated six-clause contract, with relative initiality in place of universality and an explicitly open filler on the M_6 boundary.

Two negative results are, in the author's view, the most valuable in the paper, and both are reported as failures of the finite models rather than as successes. The triadic defect on the Legendre prism decomposes (Theorem 7.17), and the finite satisfaction form fails the triadicity test (Calibration 9.12). A reader who takes the word "triadic" in the title as a claim that a non-decomposable three-fold coherence has been exhibited should take it instead as naming the programme that Problem 17.5 asks whether is realisable.

What the paper does not have is order and motion. Energy selects what is visible and not what is left or right (Problem 17.1); boundary-return words exist and periodic dynamics does not (No-go 8.21). These two absences are not oversights: they are what is left when the carriers have been built honestly and the additional structure has not been invented. Given the choice between a paper that claims a completed triadic theory and one that reports exactly where the completeness fails, this is the second.

Appendices

A Scalar, chiral, and Clifford calculations

This appendix records the computations used in §3 and §4. Nothing here is new; the purpose is to have the exact formulas in one place, so that a reader can check the paper's claims without consulting the source notes.

A.1 The three quadratic algebras

For $\kappa \in \{-1, 0, 1\}$ and $x = a + bj$, $y = c + dj$:

$$xy = (ac + \kappa bd) + (ad + bc)j, \qquad \bar{x} = a - bj, \qquad N_\kappa(x) = x\bar{x} = a^2 - \kappa b^2.$$

A.1.1 Multiplicativity of the norm

Expanding both sides,

$$N_\kappa(xy) = (ac + \kappa bd)^2 - \kappa(ad + bc)^2 = a^2c^2 + 2\kappa abcd + \kappa^2 b^2 d^2 - \kappa a^2 d^2 - 2\kappa abcd - \kappa b^2 c^2,$$

so the cross terms cancel and

$$N_\kappa(xy) = a^2c^2 - \kappa a^2 d^2 - \kappa b^2 c^2 + \kappa^2 b^2 d^2 = (a^2 - \kappa b^2)(c^2 - \kappa d^2) = N_\kappa(x) N_\kappa(y).$$

A.1.2 Inverses

When $N_\kappa(x) = a^2 - \kappa b^2 \neq 0$,

$$x^{-1} = \frac{\bar{x}}{N_\kappa(x)} = \frac{a - bj}{a^2 - \kappa b^2}.$$

The excluded loci are therefore:

$$\kappa = -1 : \ \varnothing \text{ off the origin}; \qquad \kappa = 0 : \ a = 0; \qquad \kappa = 1 : \ a = \pm b.$$

A.2 The negative-inverse involution

With $S_\kappa(z) = -1/z$ and $z = a + bj$,

$$S_\kappa(a, b) = \Big(\frac{-a}{a^2 - \kappa b^2}, \ \frac{b}{a^2 - \kappa b^2}\Big).$$

Then

$$z\, S_\kappa(z) = (a + bj) \cdot \frac{-a + bj}{a^2 - \kappa b^2} = \frac{-a^2 + abj - abj + \kappa b^2}{a^2 - \kappa b^2} = -1.$$

For $\kappa = -1$ the denominator is $a^2 + b^2$, so S_{-1} is defined on $\mathbb{R}^2 \setminus \{0\}$; and $b > 0$ gives $b/(a^2 + b^2) > 0$, so the upper half-plane is preserved. Involutivity is the standard $(S \circ S)(z) = z$ for a Möbius involution on a chart where it is defined, checked directly as

$$S_{-1}(S_{-1}(a, b)) = (a, b).$$

A.3 The exchange lift

With $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$:

$$J^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I, \qquad J^{-1} = -J.$$

Projectively, $[-\,\mathrm{id}] = [\mathrm{id}]$, hence $[J]^2 = [I]$ while $J^2 = -I$ on the lift.

On the chart $z = t/s$, the action $(t, s) \mapsto (-s, t)$ gives $z \mapsto (-s)/t = -1/z$.

A.3.1 The generator matrix

Multiplication by j in the basis $(1, j)$ is

$$L_j = \begin{pmatrix} 0 & \kappa \\ 1 & 0 \end{pmatrix}, \qquad L_j^2 = \begin{pmatrix} \kappa & 0 \\ 0 & \kappa \end{pmatrix} = \kappa I, \qquad \det L_j = -\kappa.$$

For $\kappa = -1$: $L_j = J$, det $= +1$, order four. For $\kappa = 1$: $L_j^2 = I$, det $= -1$. For $\kappa = 0$: $L_j^2 = 0$, L_j nilpotent.

A.4 Orientation-preservation of the involution

In real coordinates,

$$S(x, y) = \Big(\frac{-x}{x^2 + y^2},\ \frac{y}{x^2 + y^2}\Big).$$

As S is the holomorphic map $z \mapsto -1/z$ with derivative $1/z^2$,

$$\det DS = \left|\frac{1}{z^2}\right|^2 = \frac{1}{(x^2 + y^2)^2} > 0.$$

A.5 The six-state carrier

Let T, S, R satisfy $T^2 = S^2 = R^2 = 1$ and anticommute in pairs, and set $\Omega = TSR$.

A.5.1 Centrality

For example $T\Omega = T \cdot TSR = T^2SR = SR$ and $\Omega T = TSRT = TS(-TR) = -(TST)R = (TTS)R = SR$; the two agree. The other two generators are checked identically.

A.5.2 The square

$$\Omega^2 = T\,S\,R\,T\,S\,R = T^2\,S\,R\,S\,R = S\,R\,S\,R = -S^2R^2 = -1,$$

where the third equality moves the fourth factor past two factors (sign $+1$) and the fourth equality moves the third factor past one factor (sign -1).

A.5.3 The pairing

$$\Omega T = SR, \quad \Omega(SR) = -T, \qquad \Omega S = RT, \quad \Omega(RT) = -S, \qquad \Omega R = TS, \quad \Omega(TS) = -R,$$

so on each of the three planes $\langle T, SR\rangle$, $\langle S, RT\rangle$, and $\langle R, TS\rangle$ the operator $J = \Omega\cdot$ acts as $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

A.5.4 The characteristic polynomial

Since J acts as the same rotation on three complementary two-planes,

$$\chi_J(\lambda) = (\lambda^2 + 1)^3 = \lambda^6 + 3\lambda^4 + 3\lambda^2 + 1.$$

A.5.5 The signature trichotomy

Allowing $T^2 = \varepsilon_T$, $S^2 = \varepsilon_S$, $R^2 = \varepsilon_R$ multiplies the above reduction by $\varepsilon_T\varepsilon_S\varepsilon_R$, giving $\Omega^2 = -\varepsilon_T\varepsilon_S\varepsilon_R$. Hence:

$$\varepsilon_T\varepsilon_S\varepsilon_R = +1 \text{ (elliptic)}, \quad -1 \text{ (hyperbolic)}, \quad 0 \text{ (parabolic)}.$$

A.5.6 The chirality sign law

A permutation π of the generators acts on the word TSR by its sign, so $P_\pi \Omega P_\pi^{-1} = \operatorname{sgn}(\pi)\Omega$; in particular a transposition gives $\Omega \mapsto -\Omega = \Omega^{-1}$.

A.6 The tri-chart rotation

With

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

one computes $\operatorname{char}_R(\lambda) = \lambda^2 + \lambda + 1$, hence $R^2 + R + I = 0$ and $R^3 = I$ by Cayley–Hamilton. In coordinates: $(x, y) \mapsto (-y, x - y)$, and applying this three times returns (x, y). The three charts (x, y), $(y - x, -x)$, $(-y, x - y)$ are permuted cyclically.

B Proofs and finite counts of Part II

This appendix supplies the details omitted from Part II: the adequacy induction, the countermodels that force each hypothesis of Definition 12.3, the critical-pair configurations, the monotonicity computation behind strong normalization, and the envelope incidence count.

B.1 The adequacy induction in detail

We record the rule-by-rule obligations of Theorem 12.4.

Rule	Obligation	Discharged by
assumption	ρ realises the thread	Definition 12.2
$\wedge$I	family intersection is inhabited	hypothesis (3)
$\wedge$E	projection of an inhabited family	hypothesis (3)
$\to$I	discharged thread replaced by witness	hypothesis (1)
$\to$E	consumed thread matched	hypotheses (1), (2)
$\forall$I	a witness slot may be chosen	hypothesis (1)
$\forall$E	instantiation stays inside the family	hypothesis (3)
atom	witness or emptiness certificate	hypothesis (4)

The completeness direction uses hypothesis (4) twice: once to build the canonical assignment and once to certify emptiness when the goal family is empty. This is why (4) cannot be weakened to "either a witness or nothing found": without the certificate an empty family would be indistinguishable from an unfinished search, and by Warning 11.6 the two must not be conflated.

B.2 Countermodels for the four hypotheses

Each hypothesis is necessary for the equivalence of Theorem 12.4 in the following exact sense: there is a fixture on which the equivalence fails, the other three hypotheses hold, and the failure is attributable to the dropped hypothesis.

1. **Without (Cartesian).** Take a carrier whose slot domains are dependent, so a partial assignment may admit no extension. Then an implication whose premise is satisfiable in isolation but not jointly with the context is derivable by the rule while its conclusion has empty family.

2. **Without (Compatibility).** Take three slots with two constraints such that each pair is compatible but the triple is not. Then pairwise sections are all nonempty while the joint family is empty — the exact situation of Proposition 9.11, read as a failure of adequacy rather than as evidence for triadicity.

3. **Without (Transformer).** Take a connective with a transformer that is sound but not complete; then a derivable conclusion has nonempty family but the transformer's image is strictly larger, so family inclusion does not track derivability.

4. **Without (Exhaustiveness).** Take a carrier where some atomic family is empty with no certificate; the completeness direction fails at that atom, and the failure is reported as *unresolved* rather than as falsity.

B.3 The four critical-pair configurations

For two overlapping redexes (Definition 15.1) the ledger blocks compare as follows. Write the outer detour's ledger as $L\,[x^b]\,Q\,S$ and the inner one as sitting either inside Q or inside S.

Case	Position of the inner redex	Relation of the two reduction orders
(1)	in Q, on the discharged spine	joined after one splice step
(2)	in Q, off the spine	commute exactly
(3)	in S, adjacent block	joined after one splice step
(4)	in S, order-adjacent block	commute exactly

In cases (1) and (3) the two-step results are related by Theorem 14.3 applied to the nested substitution; the freshness condition is inherited from the outer redex. In cases (2) and (4) the blocks are disjoint, so the transports of Theorem 14.9 act on independent subterms.

B.4 Monotonicity of the two measures

Recall μ_1 counts principal-implication detours and μ_2 the total ledger size above them, compared lexicographically.

1. A root transport removes exactly one detour, so μ_1 decreases by one. The two subterms Q and S are unchanged as trees, and by Theorem 14.3 the spliced ledger has the same total size as the original block, so μ_2 is unchanged.
2. A step strictly inside a context leaves the outer detour count fixed and acts on a strictly smaller tree; induction on tree size applies. If it also removes a detour, case (1) applies.

Since $\mathbb{N} \times \mathbb{N}$ is well ordered lexicographically, no infinite reduction sequence exists, which is Theorem 14.12. Linearity enters at the first item: with contraction available, a detour could be copied and μ_1 would not decrease.

B.5 The ledger splice

For a context $L\,Q\,R$ with Q the transported middle block, the ledger of the composite is $L \circ \Lambda_\sigma \circ R$ by Theorem 14.3. Concretely, if Λ_σ replaces the i-th antecedent occurrence by a block of length ℓ_i, then the composite ledger has the same left and right blocks and a middle block of length $\sum_i \ell_i$; freshness is precisely the condition that no occurrence of the original block appears twice in the image.

B.6 Envelope incidence count

The truncated octahedron has 24 vertices, all of degree 3, and faces consisting of six squares and eight hexagons. Then

$$\#\text{face–vertex incidences} = 6 \cdot 4 + 8 \cdot 6 = 24 + 48 = 72 = 24 \cdot 3,$$

which agrees with the vertex count. The relation boundary count of Proposition 16.12 is $2m = 6$ for $m = 3$, matching Theorem 16.3.

C Six-port incidence and orbit counts

This appendix records the finite counts used in §11, §16, and Appendix B. Where a count is stated in the source material, the computation is reproduced here so that it can be checked independently.

C.1 The support partition

The carrier of Proposition 11.8 has seven nonempty faces: three vertices, three edges, and one interior. Its $2^7 = 128$ supports partition by cardinality as

$$128 = \underbrace{2}_{|S|\in\{0,7\}} + \underbrace{6}_{|S|\in\{1,6\}} + \underbrace{30}_{|S|\in\{2,5\}} + \underbrace{90}_{|S|\in\{3,4\}}.$$

The two-element classes are the empty and full supports; the six-element classes are the singletons and their complements; the remaining two classes are self-complementary in the sense that a support of size 3 has a complement of size 4.

The refinement $30 = 18 + 12$ separates the supports of size 2 and 5 into those fixed by an orientation-reversing reflection and those exchanged by it; the 18 are achiral and the 12 chiral.

C.2 The D_3 orbit count

The dihedral group D_3 of order 6 acts on the seven faces: it permutes the three vertices in a 3-cycle or a transposition, permutes the three edges identically, and fixes the interior. Burnside's lemma on the induced action on the 128 supports gives:

Element	Number	Fixed supports
identity	1	128
120° rotation	2	$2 \cdot 2 \cdot 2 = 8$ each
reflection	3	$4 \cdot 4 \cdot 2 = 32$ each

A rotation acts as a 3-cycle on the vertices and a 3-cycle on the edges, so a fixed support is determined by 2 choices per 3-cycle and 2 choices for the interior: 8. A reflection fixes one vertex and swaps a pair, and likewise for edges, so a fixed support has $2 \cdot 2$ choices on vertices and $2 \cdot 2$ on edges and 2 on the interior: 32. Hence

$$\#\text{orbits} = \frac{1}{6}\big(128 + 2 \cdot 8 + 3 \cdot 32\big) = \frac{240}{6} = 40,$$

reproducing the count of Proposition 11.8. Pairing orbits under the negation involution C_2^- halves this to the 20 archetypes stated there.

C.3 The matching count for the bridge

Theorem 16.13 uses two fixed-point-free involutions on six elements. The exact counts are as follows.

Proposition C.1 (Matching counts)**.** *On a six-element set there are*

1. 15 *fixed-point-free involutions (perfect matchings);*
2. $15 \cdot 14 = 210$ *ordered pairs of distinct such involutions;*
3. 60 *six-cycles, each admitting exactly* 2 *alternating ordered* 2*-colourings;*
4. *hence* 120 *ordered pairs whose union is connected, and* $210 - 120 = 90$ *whose union is two disjoint triangles.*

Proof. (i) A perfect matching of six elements is a partition into three unordered pairs, of which there are $\frac{6!}{2^3\,3!} = 15$. (ii) Immediate. (iii) The number of cycles on six labelled vertices up to rotation and reflection is $(6-1)!/2 = 60$, and a cycle of even length has exactly two alternating 2-colourings, one for each choice of starting colour. (iv) $60 \cdot 2 = 120$; the rest are unions of two triangles, which arise when the two matchings are the alternating classes of two disjoint 3-cycles. □

The count 120 is the exact size of the family to which Theorem 16.13 applies: connected unions are the case in which the presented boundary is a single M_6^{∂} rather than a disjoint union of smaller boundaries.

C.4 Miscellaneous finite counts

- The declared maps from six positions into three domains number $3^6 = 729$; the retention ledger of the grammar is indexed by these.
- The set partitions of six positions number $B_6 = 203$ (the sixth Bell number), which is the count of restricted-growth structures used for the allocation ledger.
- The permutohedron on four symbols has 24 vertices, matching the TO_{24} vertex count of Proposition 16.12; the two are different graphs with the same vertex count, and the paper does not identify them.
- The relation boundary count is $2m$; for $m = 3$ this is 6, which is the port count of $\mathbb{W}_6$ and the degree $\deg(\mathbb{W}_6)$ of Definition 16.8.

C.5 What these counts do and do not establish

Every count in this appendix is an exact statement about a declared finite fixture. None of them is a theorem about arbitrary carriers, and in particular:

- the 40 orbits and 20 archetypes are properties of the seven-face triangle, not of three-domain carriers in general;
- the 120 connected unions count presentations, not fillers: the filler remains open by Calibration 16.14;
- the equality of the TO_{24} and permutohedron vertex counts is a numerical coincidence unless a map is supplied, and none is supplied here.

D Claim ledger and source provenance

Every substantial result of this paper is traceable to a numbered research note of `adva/docs/research`, and the status recorded here is the status the source note assigns to itself, not an upgraded one. Where the paper states something the source does not, the entry says so.

D.1 How to read the ledger

The columns are: the paper's item; the source note; and the inherited status. The status vocabulary is that of the repository governance:

- *exact* — an algebraic or combinatorial identity proved in the paper;
- *proved* — proved in the paper under stated hypotheses;

- *calibration* — established only by a finite exhaustive check of a declared fixture;
- *proposal* — a definitional or organisational choice with no theorem;
- *no-go* — a proved negative result;
- *conjecture*, *open problem* — explicitly unproved.

D.2 Part I: observers, scalars, and the six-state carrier

Item	Source	Inherited status
Observer contract, Definition 2.1	0038	definition
Observation fibre, Definition 2.3	0038	definition
Characteristic with residual, Definition 2.4	0038	definition, strengthened by this paper to require the residual explicitly
Residual principle, Proposal 2.6	paper	proposal, derived from 0038 and the scope file
Affine product law, Lemma 2.7	0038	exact
Two-probe recovery, Proposition 2.8	0038	exact within the affine class
Interval recovery, Proposition 2.10	0038	exact; relies on the exact-interval observer
Outcome discipline, Definition 2.12	0038	methodological requirement
Lossy quotient, Proposition 2.14	0038	exact
Compression and loss, Proposition 2.15	0038	exact
Quadratic structure, Proposition 3.2	0008	exact
Three separators, Proposition 3.4	0008	exact
Negative inverse, Proposition 3.6	0008	exact
Inversion criterion, Proposal 3.7	0008	proposal: selects, does not derive
Exchange lift, Definition 3.8	0008	definition
Order two versus four, Proposition 3.9	0008	exact
Chirality, Definition 3.10	0008	definition
Not orientation reversal, Proposition 3.11	0008	exact
Generator action, Proposition 3.12	0008	exact
Bridge hypothesis, Calibration 3.13	0008	calibration
Conjugated generators, Proposition 3.15	0008	exact
Bounded trichotomy, Calibration 3.17	0008	calibration
Three readings of $6 = 2 + 2 + 2$, Proposal 3.18	0008	proposal
Six-state carrier, Definition 4.1	0009	definition
One complex unit, Proposition 4.2	0009	exact but conditional on stated anticommutation
Algebraic form is an assumption, Calibration 4.3	0009	calibration

Item	Source	Inherited status
Sign law, Proposition 4.4	0009, 0010	exact; verified for all six permutations
Signature trichotomy, Proposition 4.6	0009	exact
Temporality is not a metric sign, Warning 4.7	0009	warning
Cut interval, Construction 4.8	0010	construction
Endogenous Clifford, Theorem 4.9	0010	bounded endogenous derivation on one local three-cube
Interchange insufficient, Theorem 4.10	0010	no-go
Filtration, Proposition 4.12	0011	exact
Regrouping is not chirality, Theorem 4.13	0011	no-go

D.3 Part I: triadic inference and crossings

Item	Source	Inherited status
Typed readings, Definition 5.1	0038	definition
Variance signature, Proposition 5.2	0038 (also 0039, 0040)	proved inside the finite affine model; not a general theorem
Scope of the signature, Calibration 5.3	0038	calibration
Triadic inference, Proposition 5.4	0038	definitional identification
Deformed crossing, Proposition 5.6	0038	exact extensional law; explicitly not confluent
Square orbit quotient, Proposition 5.8	0039	exact
Two-sheeted covering, Proposition 5.9	0039	exact
Relation-valued reverse, Proposition 5.10	0039	exact
Scaling crosses, Proposition 5.12	0039	exact
Translation obstruction, Theorem 5.13	0039	no-go; forbids an *affine* crossing only
Word-to-graph boundary, Proposition 5.15	0039	structural consequence
Constructive fibre, Calibration 5.16	0039	exact finite witness
Degree profile, Definition 6.1	0040	definition
Cube degrees, Calibration 6.2	0040	calibration
Criticality warning, Warning 6.3	0040	warning
Equivariance law, Proposition 6.5	0040	exact
Equivariant characteristics, Proposal 6.6	0040	proposal
Binomial coaction, Proposition 6.8	0040	exact

Item	Source	Inherited status
Translation expands, Corollary 6.9	paper	corollary of Proposition 6.8
Isogeny three readings, Theorem 7.1	0041	external classical theorem, restated
Isogeny novelty, Calibration 7.2	0041	calibration: the bridge is prior art
Characteristics as groupoid, Proposal 7.4	0041	proposal
Three cusps, Proposition 7.5	0034	standard exact geometry
Mod-two erases chirality, Remark 7.6	0034, 0010	no-go, repaired by 0037
Oriented integer lift, Construction 7.7	0037	construction
Tri-chart coherence, Lemma 7.8	0037	lemma, direct calculation
Legendre three readings, Proposition 7.10	0042	classical local facts; the reading identification is open
Bare monodromy unipotent, No-go 7.11	0034	no-go
Central sign closure, Proposition 7.12	0042	exact
Label transport, Proposition 7.13	0042	exact
Crossing prism, Construction 7.14	0043 (prism)	construction
Crossing squares, Theorem 7.15	0043 (prism)	exact for two pairs, residual for the third
Two residual channels, Proposition 7.16	0043 (prism)	exact
Defect decomposes, Theorem 7.17	0043 (prism)	no-go; the paper's principal negative calibration
Coherence defect, Proposal 7.19	0042	proposal
Triadic crossing, Conjecture 7.20	0042	conjecture

D.4 Part I: through-crossings, logic, and the residual catalogue

Item	Source	Inherited status
Through-crossing, Definition 8.1	0047	definition
Crossing is not an arrow, Proposal 8.2	0047	proposal
Resolution torsor, Definition 8.3	0047	definition
Monodromy as difference, Theorem 8.6	0047	proved
Legendre reduction, Proposition 8.8	0047	proved from the base change
Cubical annulus, Construction 8.10	0048 (cellular)	construction
Shear transvection, Proposition 8.11	0048 (cellular)	proved
Vertex not path, Theorem 8.12	0048 (cellular)	no-go
Marked identity, Proposition 8.14	0048 (cellular)	proved
Through form, Definition 8.15	0070	definition
Order and compatibility, Proposition 8.16	0070	proved

Item	Source	Inherited status
Resolution count, Proposition 8.17	0070	exact count
Figure-eight, Theorem 8.19	0071	calibration on one word
Double periods, Proposition 8.20	0072	exact arithmetic
Return is not period, No-go 8.21	0073	no-go
Satisfaction form, Definition 9.1	0044	definition
Opposite-pair sections, Definition 9.3	0044	definition
Generated propositions, Proposal 9.4	0046	proposal
Support preorder, Definition 9.6	0044	definition
Refinement preorder, Proposition 9.7	0044	exact
Observer quotient, Calibration 9.8	0044	calibration
Adjoint quantifiers, Proposal 9.9	0046	proposal
Triadicity test, Proposal 9.11	0046	proposal, falsifiable
Triadicity fails, Calibration 9.12	0044	calibration
Three grades of logicality, Proposal 9.13	0045	proposal
Bivalence warning, Warning 9.14	0045	warning
Directional energy, Definition 9.15	0048 (energy)	definition
Energy split, Proposition 9.16	0048 (energy)	exact
Energy invariance, Proposition 9.17	0048 (energy)	exact
Zero-energy trap, No-go 9.18	0048 (energy)	no-go
Spectrum versus history, Theorem 10.3	0001, 0002	no-go
Affine spectrum, Proposition 10.4	0002	exact
Marked second jet, Theorem 10.5	0003	proved under the marked hypothesis
Co-generation, Theorem 10.6	0019	proved for the declared probe class
Observation tower, Theorem 10.8	0013	no-go
Objectification, No-go 10.9	0021	no-go
Effective one-sidedness, Proposition 10.10	0033	standard, restated
Mass is not fibre, Proposition 10.11	0033	proved
No finite coding, Proposition 10.12	0037	proved
Snapshot not empty, Proposition 10.13	0023	exact witness
Zero-event fibre, Proposition 10.14	0027	no-go
Trace sides, Proposition 10.15	0114	calibration
Count factorization, Proposition 10.16	0124	exact count plus explicit obstruction
Period not history, Proposition 10.17	0043 (prism)	no-go
Order obstruction, Theorem 10.18	0126	proved with exact minimum
Scale forgetting, Proposition 10.20	0043 (scale)	exact

Item	Source	Inherited status
Winding insufficient, No-go 10.21	0043 (path-bound)	no-go

D.5 Part II: apertures, deduction, and the kernel

Item	Source	Inherited status
Aperture-first, Proposal 11.1	0079	proposal
Typed aperture, Definition 11.2	0079	definition
Open and close, Proposal 11.3	0079	proposal
Aperture status, Definition 11.5	0079	definition
Terminal worlds, Definition 11.7	0081	definition
Seven faces, Proposition 11.8	0096	calibration
Mask direction, No-go 11.9	0096	no-go
Filling fibre, Definition 11.10	0081	definition
Fibre connectives, Proposal 11.11	0081	proposal
Surface observer, Proposal 11.13	0080	proposal
Geometric warning, Warning 11.14	0080	warning
Threaded judgment, Definition 12.1	0084	definition
Realisation, Definition 12.2	0082	definition
Adequacy hypotheses, Definition 12.3	0082	hypotheses
Adequacy, Theorem 12.4	0082	proved under the four hypotheses
Adequacy boundary, Calibration 12.8	0082	calibration
Substitution natural, Theorem 12.6	0082	proved
TND_0, Definition 13.1	0084	definition
Ledger judgment, Definition 13.2	0084	definition
Soundness, Theorem 13.3	0084	proved
Grade, Definition 13.4	0084	definition
Incompleteness, Theorem 13.5	0084	proved
Grade measures the gap, Proposition 13.6	0084	proved with exact family
Conflict aperture, Proposal 13.8	0083	proposal
Common lift, Proposition 13.9	0083	proved
One-hole context, Definition 14.1	0085	definition
Ordered substitution, Definition 14.2	0085	definition
Ledger splice, Theorem 14.3	0085	proved
Fresh cut, Corollary 14.4	0085	conditional
Raw ledger fails, No-go 14.6	0085	no-go
Ledger transport, Proposal 14.7	0086	proposal
Freshness reflects, Lemma 14.8	0086	proved
Root transport, Theorem 14.9	0086	proved
Contextual closure, Corollary 14.10	0086	proved

Item	Source	Inherited status
Two measures, Definition 14.11	0086	definition
Strong normalization, Theorem 14.12	0086	proved
Overlapping redexes, Definition 15.1	0093	definition
Four paths, Proposition 15.2	0093	proved
Substitution and graft, Lemma 15.3	0093	proved
Strong diamond, Theorem 15.4	0093	proved
Confluence, Corollary 15.5	0093	proved per fixed proof
Confluence is not coherence, No-go 15.7	0093	no-go
Normal and neutral, Definition 15.9	0094	definition
Subformula property, Theorem 15.10	0094	proved
Leftmost head, Proposition 15.11	0094	proved
Decidability, Theorem 15.12	0094	proved
Skeleton positioning, Proposition 15.13	0094	proved
No compactification, Warning 15.14	0093, 0094	warning
Whole-cut carrier, Definition 16.1	0101	definition
Six-port contract, Definition 16.2	0101	declared contract
Counting bound, Theorem 16.3	0101	proved under the contract
Six is conditional, Warning 16.5	0101	warning
The carrier $\mathbb{W}_6$, Definition 16.6	0101	definition
Line and circle as views, Proposal 16.7	0101, 0102	proposal
Whole-cut record, Definition 16.8	0102	definition
Collision conservation, Proposition 16.9	0102	proved
Cell–carrier–view, Proposal 16.11	0103	proposal
Envelope count, Proposition 16.12	0103	exact count
M_6 bridge, Theorem 16.13	0104	proved under explicit hypotheses
Filler open, Calibration 16.14	0104, 0113	calibration: no filler
Relative initiality, Proposal 16.15	0101	proposal
Not universal, Warning 16.16	0101	warning
Matching counts, Proposition C.1	paper	elementary count, check of Theorem 16.13

D.6 Source notes not absorbed as main text

The following clusters of the source corpus are mathematical but belong to neither part of this paper, and are recorded here so that the omission is visible rather than accidental.

- **0004–0007**: whole-cut program cells, the Alexandrov topology of completed causal pasts, frontier transport with its interchange square, and the occurrence-affine cut lift. This is a theory of causal cuts with its own space–time duality, presupposing Paper I's histories.
- **0015–0018**: occurrence-resolved backward probe pairing, expression-valued backward transport,

and the intensional cut-composition law. Its natural home is with Paper IV's continuation residuals.

- **0020, 0022–0031**: the finite dual-cut surgery programme, the relational and polarized PSP^* factorizations, and boundary polarization. Much of this is stated relative to checked finite artifacts and would require its own careful separation of mathematics from fixture.
- **0033, 0035–0037**: the Ω-type effective boundary, the decorated handle operator, the triangular symbolic calculus, and the finite-observer tri-cusp reduction. Only Proposition 10.10 and Proposition 10.12 are imported.
- **0050–0058**: the triadic transformation monoid, context-generated proposition algebras, causal evidence layers, and the braid/gate comparison. These are the finite models that motivate §9 but are not restated.
- **0059–0069, 0074–0078, 0087–0092, 0105–0127**: the bracket and shell calculi, the research-instrument reports, the multi-hole syntax, the venture and failure-frontier programme, and the later formation-gate notes.

Fifteen of the 133 surveyed files are predominantly engineering and contribute no main-text mathematics: `0026`, `0050` (LABS search), `0054` (sorting networks), `0074`, `0090`, `0092`, the three `0102` follow-ups, `0108`, `0110`, `0115`, `0116`, `0123`, and one SVG figure.

References

[1] Gilbert Baumslag and Donald Solitar. Some two-generator one-relator non-hopfian groups. *Bulletin of the American Mathematical Society*, 68(3):199–201, 1962.

[2] Ralph H. Fox. Free differential calculus. i. derivation in the free group ring. *Annals of Mathematics*, 57(3):547–560, 1953.

[3] Mingli Yuan. Arithmetic expression geometry 0—arithmetic expression spaces: Expansion and motion, eikonal equation, horocycle pullbacks, commutativization, contact structure, and tearing. Mathematical-review manuscript. Author, corresponding author, and sole holder of scientific responsibility: Mingli Yuan. Formalization and computational assistance by DeepSeek Harness (model `deepseek-v4-flash-vision-exp`) is credited on the title page and in the acknowledgements, not as authorship; see the repository authorship and provenance register, 2026.

[4] Mingli Yuan. Arithmetic expression geometry i: Foundations—sequential histories, projective semantics, and zero geometry. Mathematical-review manuscript, 2026.

[5] Mingli Yuan. Arithmetic expression geometry IV: Projective condensation and computational complexity. Companion manuscript, repository path `paper-4/.`, 2026.